OSIPI · GSOC 2026

Quality control for renal ASL

Extending the ASL QC ToolBox from brain to kidney — what transfers, what breaks, and what the literature will and will not let us grade.

Agnik Misra

The renal consensus specifies the acquisition, and says nothing about image quality.

Nery et al. 2020 (MAGMA, PARENCHIMA COST Action) is the renal equivalent of the 2015 ASL White Paper — 59 consensus statements. It contains **zero numeric image-quality thresholds**: no tSNR cut-off, no CoV cut-off, no motion limit, no QEI equivalent.

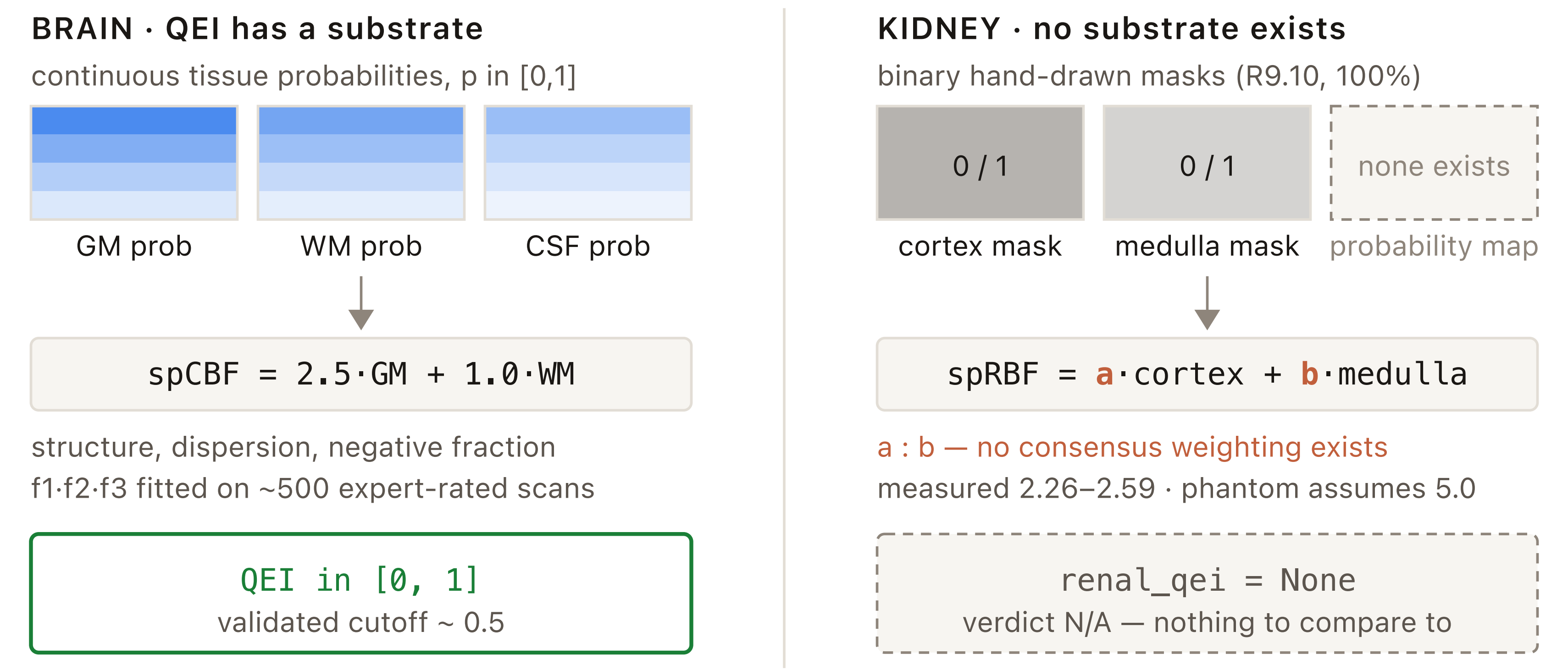
So Stream A — was the scan acquired correctly — is richly evidenced and directly checkable. Stream B — is the perfusion map good — has no published thresholds at all, so every number we set there ships **UNCALIBRATED** and the report says so.

> Two rules do transfer, and both are honoured: **R10.1** (100% agreement) report cortical RBF, not whole-kidney, separately for left and right. **R10.2** (89%) medullary values are not considered reliable — which is why the cortex:medulla ratio is a segmentation-integrity flag, never a perfusion verdict.

MODULE K1

Module K1 — Quality index

The QEI slot in the renal toolbox — registered, always N/A, and the four reasons it ships empty.

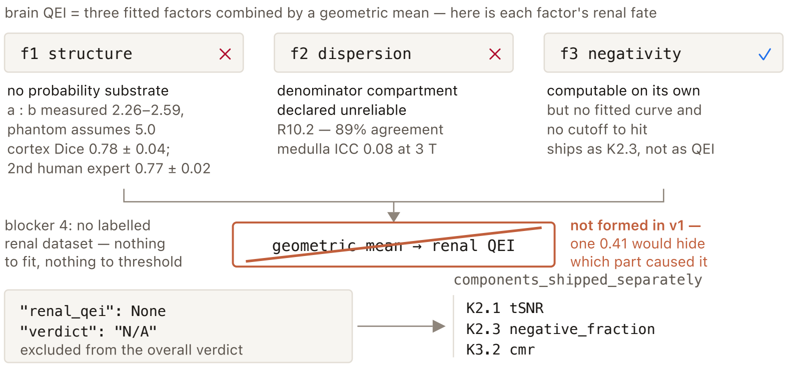


K1.1 is registered and always N/A — the empty slot is visible in every report.

K1.1 — Renal composite quality index · REGISTERED, ALWAYS N/A IN v1

This check asks whether there is a single 0-to-1 number summarising renal perfusion-map quality the way QEI does for brain — and in version one it answers no, out loud, in the report.

It stops the absence of a renal anchor metric from being an unexplained gap, so no reader mistakes a Stream-B verdict for one that had a quality index behind it.



1. Accept whatever was supplied — RBF map, cortex masks, medulla masks — and consume none of it; the answer does not depend on the inputs.
2. Return N/A unconditionally, with the four blockers attached. No arithmetic runs at all.
3. Emit the ids of the checks that carry the individually-computable pieces, so the reader is routed to what does exist instead of seeing an unexplained gap.
4. Do NOT form a composite from those pieces — that is a decision, not a deferral: a geometric mean of three uncalibrated components is less interpretable than its components, and none of the three has a validated scale to be averaged on.

INPUT

```
{
  "rbf_map":      "NIfTI 3D float | None",
  "cortex_masks": "{'left': NIfTI 3D bool, 'right': NIfTI 3D bool} | None",
  "medulla_masks":"{'left': NIfTI 3D bool, 'right': NIfTI 3D bool} | None",
}
```

OUTPUT

```
{
  "metric": {
    "renal_qei": None,
    "blockers": ["no_probability_substrate",
                 "boundary_not_reliably_drawable",
                 "denominator_compartment_declared_unreliable",
                 "no_labelled_dataset"],
    "components_shipped_separately": ["K3.2.cmr",
                 "K2.3.negative_fraction", "K2.1.tsnr"],
  },
  ... 4 more lines
}
```

VERDICT

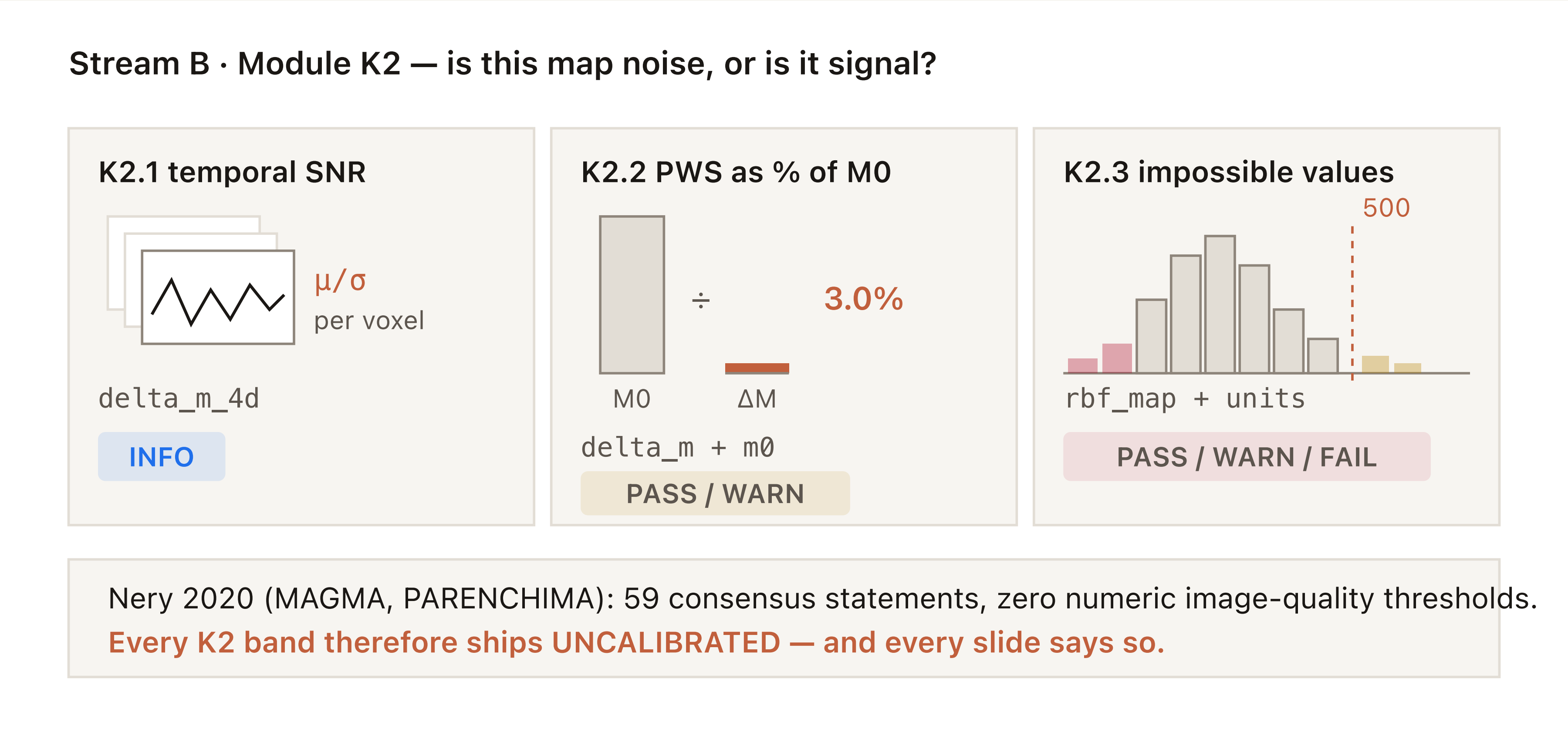
`N/A — excluded from the overall verdict, so the empty slot cannot drag a scan down` always in v1

UNCALIBRATED — there is nothing to cite, and the absence IS the finding: Nery 2020 term scan returns zero hits for quality control, CoV or cut-off; blocker 3 rests on R10.2 (89% agreement, 14% abstentions).

> The question Dolui will ask is "why not just build one?" — the honest answer is blocker 4: QEI's f1 f2 f3 and its ~0.5 cutoff were fitted against expert ratings on ~500 scans, and no renal ASL dataset with expert quality ratings exists. Without labels there is nothing to fit and nothing to threshold; a QEI-Net analogue is further out still, for the same reason.

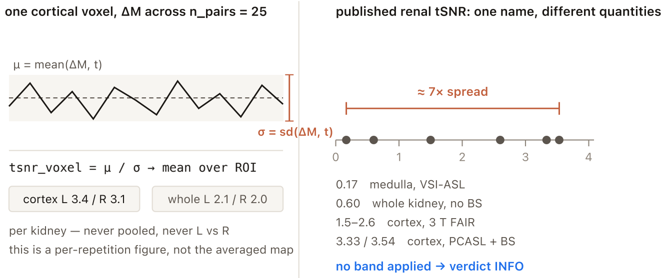
Module K2 — Noise & distribution

Three checks on the renal perfusion map: how stable the subtraction is, how big it is relative to M0, and how much of it is physically impossible.



K2.1 Temporal SNR (tSNR) — REQUIRED, report-only

How stable is the perfusion-weighted signal across repetitions?



1. Drop non-finite volumes and record n_pairs_used per kidney — it can differ per side if K7.2 already rejected pairs on one side only.
2. Voxelwise mu = numpy.mean(dM, axis=3) and sd = numpy.std(dM, axis=3, ddof=1) over the repetition axis.
3. tsnr_voxel = mu / sd, excluding sd <= 0 voxels, then average over the ROI — cortex when a cortex mask exists, whole kidney otherwise.
4. Emit definition and per_repetition: True inside the metric, with the full acquisition context, because renal tSNR is not comparable across papers otherwise.
5. Report per kidney. Never pool, never compare left against right. Then emit INFO and stop — no band is applied.

INPUT

```
{
  "delta_m_4d":   "NIfTI 4D float",   # (X, Y, Z, n_pairs) per-repetition C-L subtractions
  "kidney_masks": {"left": "NIfTI 3D bool", "right": "NIfTI 3D bool"},
  "cortex_masks": {"left": "NIfTI 3D bool", "right": "NIfTI 3D bool"},  # optional, preferred
  "context": {"labelling": "FAIR", "field_T": 3.0,
              "readout": "2D SE-EPI", "n_pairs": 25},
}
```

OUTPUT

```
{
  "metric": {
    "left":  {"tsnr_cortex": 3.4, "tsnr_whole": 2.1, "n_pairs_used": 24, "n_voxels": 430},
    "right": {"tsnr_cortex": 3.1, "tsnr_whole": 2.0, "n_pairs_used": 23, "n_voxels": 418},
    "definition":     "voxelwise mean(dM, t) / sd(dM, t), then averaged over ROI voxels",
    "per_repetition": True,          # NOT the tSNR of the averaged map
    "context":    {"labelling": "FAIR", "field_T": 3.0, "readout": "2D SE-EPI", "n_pairs": 25},
    "provenance": "uncalibrated - no renal tSNR cutoff exists; reported for inspection only",
  },
  "verdict": "INFO | UNKNOWN",
  ... 2 more lines
}
```

VERDICT

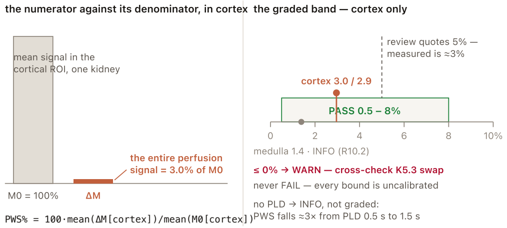
INFO	always, when it runs — the numbers and their context are reported, never graded
UNKNOWN	no 4D series (a single averaged map has no temporal axis), no mask, or fewer than 3 usable repetitions

UNCALIBRATED — no renal tSNR cutoff exists. Definition from harteveld2020 and garciaruiz2025. Reference context only: cortex 3.54 ± 0.71 at 1.5 T, 3.33 ± 0.54 at 3 T (garciaruiz2025, n=16); cortex 1.5–2.6 (buchanan2018, n=10); whole kidney 0.60 ± 0.15 rising to 0.93 ± 0.22 with heavy BS (bones2019, n=9); medulla 0.17 ± 0.14 (franklin2021). Per-kidney reporting mandated by R10.1 (0% abstentions, 100% agreement).

No band is set, for two specific reasons. (1) The definitions are not the same quantity: Buchanan's tSNR divides by the SD across the 25 ASL pairs — per-repetition — while an averaged map built from 25 pairs has roughly sqrt(25) = 5x higher SNR, so a copied threshold would be about 5x too strict and would fail nearly every good kidney scan. (2) Published values span about 7x (0.17 to 3.54), driven by ROI, field strength, labelling, readout and repetition count — not by data quality. It never falls back to a whole-image tSNR: outside the kidney the denominator is noise.

K2.2 Perfusion-weighted signal as % of M0 — REQUIRED

Is the ASL difference signal a physically sensible magnitude relative to the calibration image — the renal analogue of "does this look like ~1% ASL contrast?"



1. Gate on space first: if K4.2 reports ΔM and M0 are not on the same grid and no transform was supplied, return UNKNOWN — a ratio across mismatched grids is a number about interpolation, not labelling.
2. For each kidney and each available ROI: PWS% = 100 · mean(delta_m[roi]) / mean(m0[roi]), using numpy.nanmean on both.
3. If mean(m0[roi]) <= 0, return UNKNOWN for that kidney — the denominator is not a PD signal.
4. Grade the cortical value only against the 0.5–8% band. Medulla is INFO (R10.2); whole-kidney is INFO (R10.1 asks for cortex).
5. If pld_or_ti_s is absent, compute and report but demote to INFO — PWS falls by roughly 3× between PLD 0.5 s and 1.5 s in the same subjects. Take the worse of the two kidneys and name the side.

INPUT

```
{
  "delta_m":      "NIfTI 3D float",   # the mean subtraction across pairs
  "m0":           "NIfTI 3D float",   # calibration / PD image, same grid
  "kidney_masks": {"left": "NIfTI 3D bool", "right": "NIfTI 3D bool"},
  "cortex_masks": {"left": "NIfTI 3D bool", "right": "NIfTI 3D bool"},  # optional, preferred
  "medulla_masks":{"left": "NIfTI 3D bool", "right": "NIfTI 3D bool"},  # optional, INFO only
  "pld_or_ti_s":  1.4,               # required to interpret; None -> reported, not graded
  "n_bs_pulses":  3,                 # optional; BS scales dM and M0 differently
}
```

OUTPUT

```
{
  "metric": {
    "left":  {"pws_cortex_pct": 3.0, "pws_medulla_pct": 1.4, "pws_whole_pct": 2.4},
    "right": {"pws_cortex_pct": 2.9, "pws_medulla_pct": 1.3, "pws_whole_pct": 2.3},
    "pld_or_ti_s": 1.4,
    "band_used":   [0.5, 8.0],
    "provenance":  "uncalibrated band drawn to contain the full published spread",
  },
  "verdict": "PASS | WARN | UNKNOWN",
  "reason":  "cortical PWS 3.0% / 2.9% at PLD 1.4 s – within the reported renal range",
}
```

VERDICT

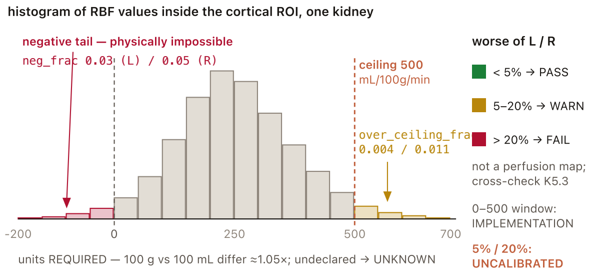
PASS	cortical PWS within 0.5–8% on both kidneys
WARN	cortical PWS outside 0.5–8% on either kidney — labelling weak, or M0 mis-scaled
WARN	cortical PWS ≤ 0 on either kidney — points at a subtraction-sign failure; cross-check K5.3
FAIL	never — every bound here is uncalibrated, and a low PWS at a long PLD is a legitimate physiological finding, not a defect
INFO	computed but pld_or_ti_s unknown, so the band cannot be applied honestly
UNKNOWN	no M0, no mask, ΔM and M0 not co-registered, or mean(M0[roi]) ≤ 0

The 0.5–8% band is UNCALIBRATED — drawn to contain the full published spread, not fitted. References: cortex 2.95 ± 0.56% (1.5 T) and 3.09 ± 0.59% (3 T), medulla 1.36 ± 0.15% and 1.40 ± 0.19% (garciaruiz2025, n=16); whole kidney 3.43 ± 1.43% (radovic2022); cortex peak 5.98 ± 0.70% (bones2021); paediatric 1.5 ± 0.61% at PLD 0.5 s falling to 0.51 ± 0.27% at PLD 1.5 s (harteveld2022); review statement "typical renal cortex perfusion-weighted signal intensity of 5% of the control image signal" (odudu2018). Measured values sit below the quoted 5%. Cortex-only grading follows R10.2 (14% abstentions, 89% agreement); medulla and whole-kidney are INFO (R10.1, 100% agreement).

Catches labelling failure, wrong subtraction sign, M0 scaling errors, and background suppression far more or less aggressive than declared. n_bs_pulses is recorded with the metric and a BS-heavy acquisition is annotated rather than re-banded — there is no published BS-conditioned PWS band to re-band onto.

K2.3 Implausible-value and negative-voxel fraction — REQUIRED

What fraction of within-kidney voxels are physiologically impossible?



1. Unit guard before any arithmetic: the published clip is stated per 100 g at origin and re-cited per 100 mL elsewhere — different normalisations, ~1.05 g/mL apart. If units is not declared, return UNKNOWN; do not guess.
2. Select the graded ROI: cortex where a cortex mask exists, whole kidney otherwise, and record which in roi.
3. neg_frac = numpy.count_nonzero(rbf[roi] < 0) / rbf[roi].size
4. over_ceiling_frac = numpy.count_nonzero(rbf[roi] > 500) / rbf[roi].size
5. Score each kidney; the verdict is the worse of the two, naming the side. Do not clip the map — this is QC: it reports how much of the map is impossible, repair is the quantification pipeline's job.

INPUT

```
{
  "rbf_map":      "NIfTI 3D float",
  "kidney_masks": {"left": "NIfTI 3D bool", "right": "NIfTI 3D bool"},
  "cortex_masks": {"left": "NIfTI 3D bool", "right": "NIfTI 3D bool"},  # optional, preferred
  "units":        "mL/100g/min",    # or "mL/100mL/min"; REQUIRED — no default is assumed
}
```

OUTPUT

```
{
  "metric": {
    "left":  {"neg_frac": 0.03, "over_ceiling_frac": 0.004, "n_voxels": 812, "roi": "cortex"},
    "right": {"neg_frac": 0.05, "over_ceiling_frac": 0.011, "n_voxels": 790, "roi": "cortex"},
    "ceiling": 500.0,
    "units":   "mL/100g/min",
  },
  "verdict": "PASS | WARN | FAIL | UNKNOWN",
  "reason":  "3% (L) / 5% (R) negative cortical voxels; ceiling exceedance under 2%",
}
```

VERDICT

PASS	negative fraction < 5% and ceiling-exceedance fraction < 5%, on both kidneys
WARN	either fraction in 5–20% on either kidney
FAIL	negative fraction > 20% on either kidney — a majority-negative perfusion map is not a perfusion map; cross-check K5.3 for a control/label inversion
UNKNOWN	no mask, units not declared, or fewer than 20 ROI voxels

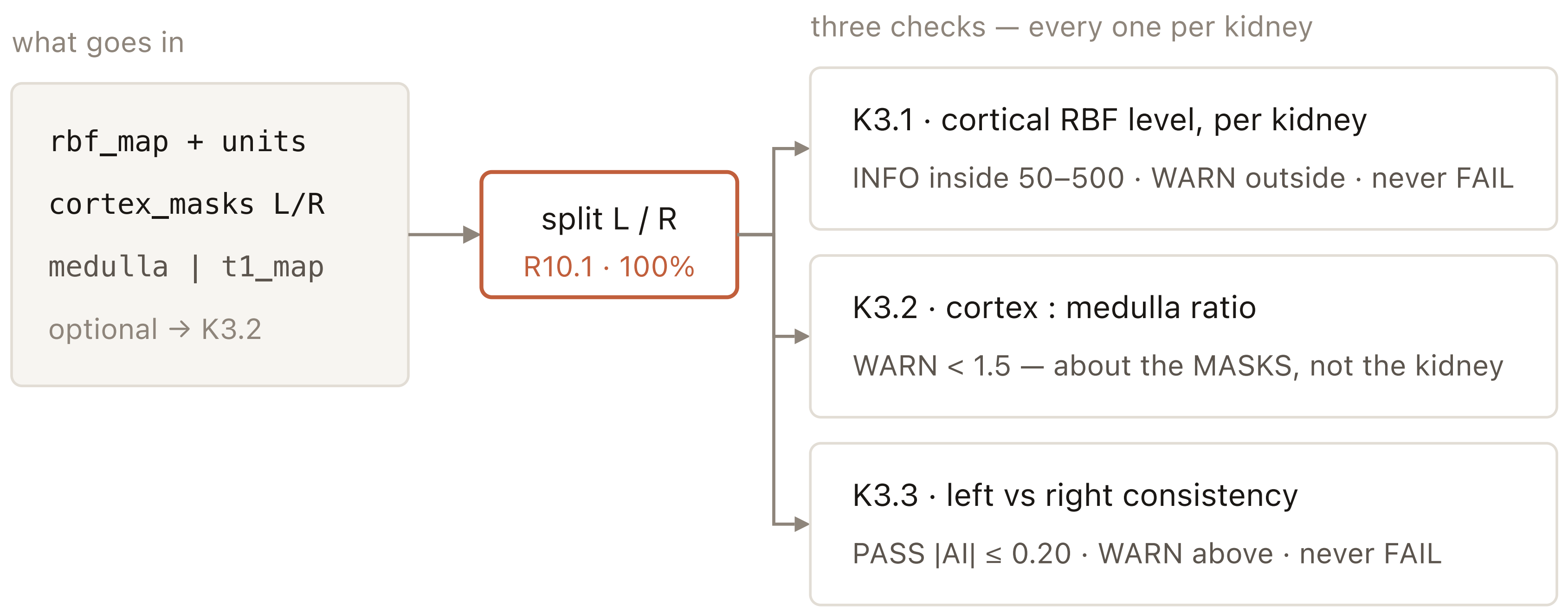
The 0–500 voxel window is IMPLEMENTATION — real and published (Gullaksen 2023, applied in olsen2025), but a per-voxel preprocessing clip inside one study's segmentation step, not a validated scan-level quality bound. The 5% / 20% cut-points are UNCALIBRATED: nobody publishes "reject the scan if X% of cortical voxels are negative"; they are carried from the brain module (10%/20%) and tightened to 5% because renal cortical ROIs are small. Ceiling plausibility context: whole-kidney flow ~1200 mL/min ≈ 400 mL/100 g/min in a 70 kg adult with a 300 g kidney (odudu2018). Per-kidney reporting: R10.1 (0% abstentions, 100% agreement).

This is the only Stream-B FAIL that grades perfusion values at all — the other two (K4.1 empty or overlapping masks, K4.2 grid mismatch) are structural. The FAIL is licensed by the sign, not by the 20%: negative perfusion is impossible, and the 20% only sets how much noise-driven negativity is tolerated. A reader unwilling to let an uncalibrated fraction gate a FAIL can set strict=False, which demotes it to WARN. On a raw ΔM rather than a quantified map, K5.2's structure field gates it: the negative half runs, the ceiling half reports None. Catches control/label swap, subtraction sign errors, vascular contamination and gross quantification-constant errors.

Module K3 — Perfusion level & cortico-medullary contrast

K3

The three checks that grade the perfusion number itself — cortical level per kidney, cortex-vs-medulla separation, left-vs-right consistency. Every threshold here ships UNCALIBRATED, so no K3 check can FAIL.

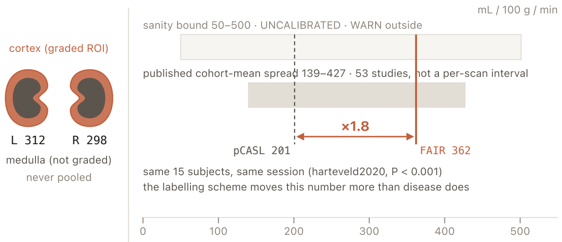


No K3 check can FAIL — every threshold here is UNCALIBRATED (Nery 2020 states no numeric quality bound)

K3.1 · Cortical RBF level, per kidney

K3.1 asks the one question the renal consensus explicitly instructs us to report on: is cortical renal blood flow, for each kidney separately, a physiologically possible number?

Catches gross quantification errors, wrong constants, wrong units, and a mask placed outside the kidney — without ever pretending to detect abnormal perfusion.



R10.1 (0% abst / 100%): report cortical RBF, left and right separately

1. Refuse to run without a declared `units` field → UNKNOWN; per 100 g and per 100 mL differ by the ~1.05 g/mL tissue density.
2. Per kidney, over the cortex mask: nanmean, nanstd(ddof=1) and nanmedian — the median because the consensus explicitly asks for it in skewed RBF distributions.
3. Also compute the whole-kidney mean and emit it clearly labelled as NOT the consensus quantity, so a user with only whole-kidney masks still gets a number.
4. Report left and right separately; no pooled mean is emitted at all, so nothing downstream can accidentally use one (R10.1).
5. Grade only against the wide 50–500 sanity band: inside → INFO with full acquisition context attached; outside → WARN. No low-side FAIL — a thin cortex partial-volumes downward, which is anatomy, not data quality.

INPUT


```
{
  "rbf_map":      "NIfTI 3D float",
  "cortex_masks": {"left": "NIfTI 3D bool", "right": "NIfTI 3D bool"},
  "kidney_masks": {"left": "NIfTI 3D bool", "right": "NIfTI 3D bool"},   # fallback ROI, reported only
  "units":        "mL/100g/min",       # REQUIRED
  "context":      {"labelling": "FAIR", "field_T": 3.0, "pld_or_ti_s": 1.9,
                   "subject_age": 51, "native_or_transplant": "native",
                   "lambda": 0.9, "alpha": 0.95, "t1_blood_s": 1.65},
}
```


OUTPUT


```
{
  "metric": {
    "left":  {"cortical_rbf_mean": 312.0, "cortical_rbf_sd": 48.0,
              "cortical_rbf_median": 305.0, "n_voxels": 430},
    "right": {"cortical_rbf_mean": 298.0, "cortical_rbf_sd": 51.0,
              "cortical_rbf_median": 291.0, "n_voxels": 418},
    "whole_kidney_mean": {"left": 241.0, "right": 236.0},   # reported, NOT the consensus quantity
    "sanity_band": [50.0, 500.0],
    "band_used":   "none - no per-scan reference interval exists; sanity bound only",
  },
  ... 3 more lines
}
```


VERDICT

ℹ️ INFO	cortical mean within 50–500 — value reported with its acquisition context, no quality claim made
⚠️ WARN	cortical mean outside 50–500 mL/100 g/min on either kidney — a sanity bound, not a normal range
❌ FAIL	never — no per-scan reference interval exists
⊘ N/A	no cortex mask but a whole-kidney mask exists — the whole-kidney mean is reported and explicitly not graded
? UNKNOWN	no mask at all, `units` undeclared, or fewer than 20 cortical voxels

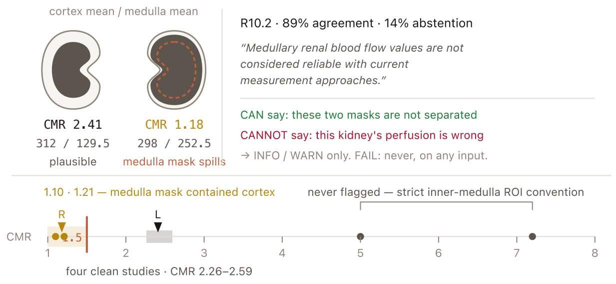
R10.1 (0% abst / 100%) mandates the quantity; the 50–500 band itself is UNCALIBRATED — 139–427 is a spread of cohort means across 53 studies (odudu2018, p.2), not a per-scan reference interval.

Why no FAIL, and no tighter band? Technical factors move this number more than disease does. In the same 15 subjects, same session, FAIR gave 362 ± 57 vs pCASL 201 ± 72 mL/min/100 g ($P < 0.001$, harteveld2020) — larger than any healthy-vs-diseased or adult-vs-child difference in this entire evidence base. Field strength moves it ~11%, age ~20%, ROI convention ~7%, quantification constants ~22%. And the PET reference standard itself disagrees with ASL by limits of agreement of ±136 mL/min/100 mL (olsen2025). There is also no cross-centre reproducibility study at all (odudu2018, p.4).

K3.2 · Cortico-medullary ratio (CMR) — a segmentation-integrity flag

K3.2 divides mean cortical RBF by mean medullary RBF per kidney — and because the consensus itself declares medullary values unreliable, that ratio is allowed to say the two masks are drawing from the same tissue, and nothing more than that.

Catches cortex and medulla masks that are not actually separated — the dominant renal segmentation failure, with a published worked example in Hammon's own admission that "the segmentation of the medulla may contain parts of the cortex".



1. If no medulla mask is supplied but a quantitative T1 map is, derive one inside the kidney mask: medulla is the high-T1 compartment, split at 1247 ms at 3 T or 1067 ms at 1.5 T; tag mask_source "derived_from_t1" and mark the verdict provisional.
2. Per kidney: cmr = nanmean(rbf[cortex]) / nanmean(rbf[medulla]).
3. Guard the denominator — if medulla_mean ≤ 0, or the medulla mask has fewer than 10 voxels, return UNKNOWN for that kidney rather than dividing.
4. Emit both compartment means alongside the ratio, so a reader can see which side is anomalous: a CMR of 1.2 from a low cortex is a different problem from one from a high medulla.
5. Interpret only in the low direction: cmr < 1.5 → WARN, "cortex and medulla masks may not be separated". High CMR is never flagged, and this check never FAILs or escalates beyond a WARN about masks.

INPUT

```
{
  "rbf_map":       "NIfTI 3D float",
  "cortex_masks":  {"left": "NIfTI 3D bool", "right": "NIfTI 3D bool"},
  "medulla_masks": {"left": "NIfTI 3D bool", "right": "NIfTI 3D bool"},   # or derived from t1_map
  "t1_map":        "NIfTI 3D float | None",    # ms; used to derive medulla when no mask is supplied
  "field_T":       3.0,                        # selects the T1 split point
}
```

OUTPUT

```
{
  "metric": {
    "left":  {"cmr": 2.41, "cortex_mean": 312.0, "medulla_mean": 129.5,
              "mask_source": "supplied"},
    "right": {"cmr": 1.18, "cortex_mean": 298.0, "medulla_mean": 252.5,
              "mask_source": "supplied"},
    "flag":  {"left": "plausible", "right": "compartments_likely_not_separated"},
    "trip_point": 1.5,
  },
  "verdict": "INFO | WARN | UNKNOWN | N/A",
  ... 2 more lines
}
```

VERDICT

- **INFO** — CMR ≥ 1.5 on both kidneys — reported, no quality claim made
- **WARN** — CMR < 1.5 on either kidney → "cortex and medulla masks may not be separated" — a statement about the masks, not about the kidney
- **FAIL** — never, on any input
- **N/A** — no medulla mask and no T1 map — the consensus default protocol produces cortex ROIs only, so this is the common case and must not be penalised
- **UNKNOWN** — medulla mean ≤ 0, or either mask has too few voxels to average

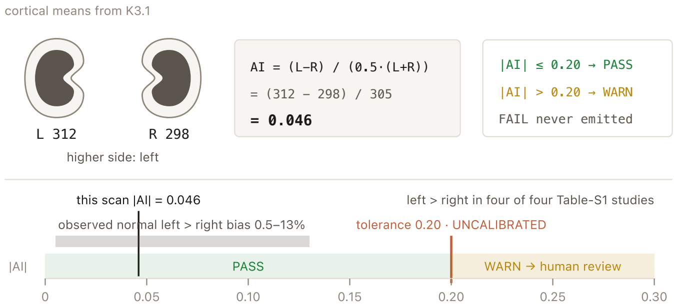
> R10.2 (14% abst / 89%) is what demotes it to a mask flag; the 1.5 trip point is UNCALIBRATED — no paper states a cortex:medulla ratio threshold for any purpose.

> This is the brain GM/WM ratio in form — scale-free, robust to global scaling errors — but deliberately not in role. In brain, GM/WM < 1 is a published data-quality FAIL. Here R10.2 makes the denominator unreliable, so the ratio may only speak about the ROIs. Two supports: only ~10% of renal blood flow perfuses the medulla, so a CMR near 1 is anatomically impossible; and CMR is not even monotone with disease — it rises 1.57 → 2.01 in one CKD cohort and falls 2.08 → 1.75 in another. It is also the one Stream-B check a global scaling error cannot fool, which is exactly why it is worth having alongside K3.1.

K3.3 · Left-vs-right consistency

K3.3 compares the two kidneys against each other, because a one-sided failure of masks, registration or motion correction is invisible to every per-kidney check — each side looks perfectly consistent on its own.

Catches one-sided mask, registration or motion-correction failure — the failure mode invisible to every per-kidney check. Genuinely asymmetric pathology is surfaced for review rather than graded.



1. Take the two cortical means from K3.1. If either is UNKNOWN, this check is UNKNOWN — it never substitutes a whole-kidney mean, which would mix an ROI effect with a physiological one.
2. Compute AI = (L − R) / (0.5·(L + R)), the standard normalised-difference definition, identical to the brain asymmetry check.
3. Compare abs(AI) against 0.20.
4. Report which side is higher, because the expected normal bias has a direction — left > right — so a right-higher asymmetry of the same size is marginally more interesting.
5. Never FAIL, and grade perfusion only: arterial transit time does differ significantly between kidneys with FAIR even when perfusion does not, so an ATT asymmetry check would fire on normal physiology.

INPUT

```
{
  "cortical_rbf": {"left": 312.0, "right": 298.0},   # means from K3.1, same units
  "n_kidneys":    2,                                  # 1 for transplant / nephrectomy / agenesis
  "units":        "mL/100g/min",
}
```

OUTPUT

```
{
  "metric": {
    "asymmetry_index": 0.046,     # (L − R) / (0.5 * (L + R))
    "left": 312.0, "right": 298.0,
    "tolerance": 0.20,
    "higher_side": "left",
  },
  "verdict": "PASS | WARN | UNKNOWN | N/A",
  "reason":  "left-right asymmetry 4.6% — within the tolerated normal bias",
}
```

VERDICT

✅ PASS	|AI| ≤ 0.20
⚠ WARN	|AI| > 0.20 → flag for human review; may be one-sided mask/registration failure or real unilateral pathology
❌ FAIL	never — asymmetry is a review flag, not a rejection
⊘ N/A	n_kidneys == 1 (transplant, nephrectomy, agenesis)
? UNKNOWN	either kidney's cortical RBF is unavailable

UNCALIBRATED — 0.20 is set above the largest observed normal left>right bias (~9–13% across three PET/MR sessions) and comfortably above the cortical between-visit CV of 4–13% (odudu2018, p.3).

Why tolerate as much as 20%? Four independent cohorts find no significant left–right perfusion difference (P = 0.93 FAIR, P = 0.52 pCASL, within-subject r = 0.83/0.94), but a consistent small left > right bias does exist — 0.5–6.8% in four Table-S1 studies, left higher in four of four, and 9–13% across three PET/MR sessions. A tolerance tighter than ~13% would fire on normal anatomy. And a genuinely large asymmetry may be renal artery stenosis or a single functioning kidney: a clinical finding, not an image defect.

MODULE K4

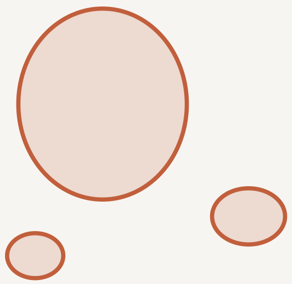
K4

Module K4 — Masks & co-registration

Three REQUIRED structural gates: is the mask sane, is everything in one space, and how many slices survived?

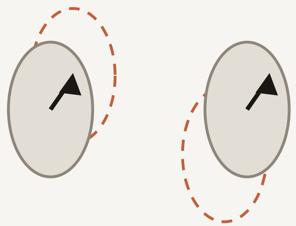
Every Stream B number is a mean over a mask, in a space, across slices. K4 tests all three.

`K4.1 mask integrity`



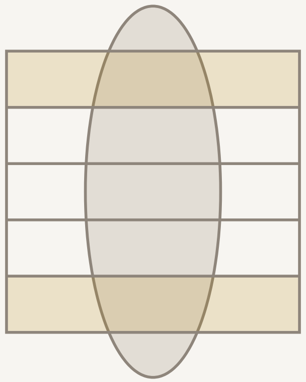
`n_components: 3`
`largest_component_frac`
`= 0.71 → WARN`

`K4.2 one space`



`scope: "global"`
one rigid transform,
two organs → WARN

`K4.3 slice coverage`



`usable_fraction 0.60`
stack edges are lost
first → WARN

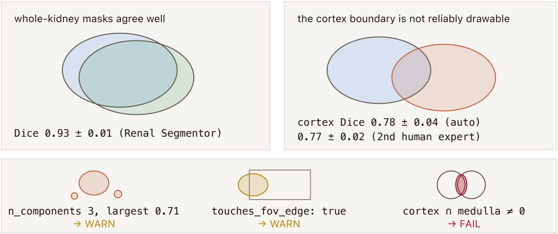
All three REQUIRED: if K4 is wrong, every K1–K3 number is a mean over the wrong voxels.

K4.1 Per-kidney mask integrity

Are the supplied masks structurally sane — one connected component per kidney, non-empty, non-overlapping, inside the field of view, and thick enough to support a cortical mean?

Catches the mask problems that silently corrupt every other Stream-B number: a mask that caught bowel or liver as a second component, a kidney cut off by the FOV, cortex and medulla ROIs drawn overlapping, and a cortex too thin for the voxel size to resolve.

Structural sanity first — and remember how well even a good mask agrees.



1. Connected components in pure NumPy (osipy bans scipy): iterative flood fill, 6-connectivity. Report n_components and largest_component_frac — the fraction matters more than the count, because a 3-voxel speckle is a different problem from a mask split in half.
2. Count n_voxels. An empty mask is a FAIL, not UNKNOWN: a mask was supplied and it does not describe a kidney.
3. FOV-edge test over all six faces — mask[0,:,:].any() or mask[-1,:,:].any() … A kidney clipped by the FOV means the reported mean is over a truncated organ.
4. Overlap tests: logical_and(cortex, medulla) per kidney and logical_and(left, right). Any non-zero is a construction error, not a borderline measurement — the compartments are defined disjoint.
5. Cortical thickness estimate: thickness_vox ≈ cortex_volume / cortex_surface, surface counted as cortex voxel faces adjacent to non-cortex (shifted-array comparison, three axes), converted with voxel_size_mm. Verdict = the worse kidney, named by side.

INPUT

```
{
  "kidney_masks":  {"left": "NIfTI 3D bool", "right": "NIfTI 3D bool"},
  "cortex_masks":  {"left": "NIfTI 3D bool", "right": "NIfTI 3D bool"},   # optional
  "medulla_masks": {"left": "NIfTI 3D bool", "right": "NIfTI 3D bool"},   # optional
  "voxel_size_mm": [3.0, 3.0, 5.0],
  "image_shape":   [96, 96, 5],
}
```

OUTPUT

```
{
  "metric": {
    "left":  {"n_voxels": 812, "n_components": 1, "largest_component_frac": 1.00,
              "touches_fov_edge": False, "cortex_voxels": 430,
              "cortex_thickness_mm_est": 6.3, "cortex_thickness_vox_est": 2.1},
    "right": {"n_voxels": 790, "n_components": 3, "largest_component_frac": 0.71,
              "touches_fov_edge": True,  "cortex_voxels": 402,
              "cortex_thickness_mm_est": 5.9, "cortex_thickness_vox_est": 2.0},
    "cortex_medulla_overlap_voxels": {"left": 0, "right": 0},
    "kidney_kidney_overlap_voxels": 0,
  ... 4 more lines
}
```

VERDICT

PASS	one dominant component per kidney (largest ≥ 95% of voxels), no overlap, not FOV-clipped, ≥ 50 cortical voxels, estimated cortical thickness ≥ 2 voxels
WARN	fragmented (largest component < 95%), FOV-clipped, fewer than 50 cortical voxels, or estimated cortical thickness < 2 voxels
FAIL	an empty mask, or cortex ∩ medulla non-empty, or left ∩ right non-empty
UNKNOWN	no masks supplied

Structural rules (components, overlap, edge contact, empty) are definitions — nothing to tune. The ≥ 2-voxel cortical-thickness line is UNCALIBRATED, but the mechanism is consensus-published: "the reduction in cortical thickness can be severe enough such that it approaches the typical dimensions of the ASL voxels … cortical RBF results from ASL should be interpreted with caution in cases where cortical thinning is evident", with recommended in-plane resolution 2–4 mm (R6.10, 0% abst / 93%). The ≥ 50 cortical voxels floor is UNCALIBRATED — shi2026 used one cortical ROI of 70–110 voxels, so 50 sits below normal practice.

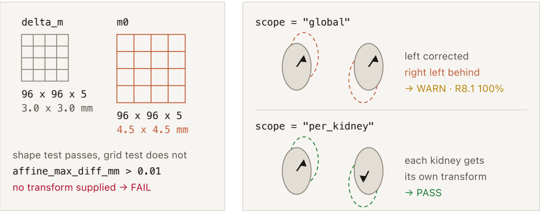
Needs at least whole-kidney masks per side plus voxel size. Cortex/medulla tests are skipped (not failed) when those masks are absent; no masks at all → UNKNOWN. This is also why K3.1 carries no low-side FAIL: a low cortical RBF can be an artefact of anatomy meeting voxel size.

K4.2 ASL ↔ M0 and per-kidney registration

Are the perfusion series, the M0 and the masks actually in the same space, and was registration done per kidney?

Catches masks applied to the wrong voxels, an M0 that cannot be used as a per-voxel denominator, and the specifically renal failure of correcting both kidneys with one transform.

Same shape is not the same grid — and two kidneys need two transforms.



residual_centroid_shift_mm: left 0.8 · right 1.1 — WARN above 1 in-plane voxel (3.0 mm)

1. Compare grids, not just matrix sizes: shapes must match on the first three dimensions AND affine_max_diff_mm = numpy.abs(A_m0 - A_asl).max() over the 3×4 geometry block must sit under a 0.01 mm float-round-trip tolerance. Two images can share 96×96×5 on different voxel sizes.
2. If the grids differ and no transform is supplied → FAIL: every ROI statistic would be computed over the wrong voxels, so this is a corruption, not an absence. If a transform is supplied, PASS on geometry and record it.
3. Read registration.scope. scope == "global" → WARN: a single rigid transform for both kidneys cannot be right, because they move independently.
4. Residual check when masks and the 4D series are both present: intensity-weighted centroid of each kidney mask region in the first and last volumes, distance reported in mm. Cheap, dependency-free, and it feeds K7.1 which grades displacement properly.
5. Verdict is the worst of: grid test, scope test, and the residual shift exceeding one in-plane voxel.

INPUT

```
{
  "delta_m_affine": "4x4 float", "delta_m_shape": (96, 96, 5, 25),
  "m0_affine":      "4x4 float", "m0_shape":      (96, 96, 5),
  "mask_affine":    "4x4 float", "mask_shape":    (96, 96, 5),
  "registration": {                   # provenance, as recorded upstream
     "performed": True,
     "scope":     "per_kidney",        # "per_kidney" | "global" | "per_kidney_per_slice" | None
     "n_transforms": 2,
     "type":      "rigid",
  },
  ... 2 more lines
}
```

OUTPUT

```
{
  "metric": {
    "grids_match":        {"m0_vs_asl": True, "mask_vs_asl": True},
    "affine_max_diff_mm": 0.0,
    "registration_scope": "per_kidney",
    "residual_centroid_shift_mm": {"left": 0.8, "right": 1.1},
    "voxel_size_mm": [3.0, 3.0, 5.0],
  },
  "verdict": "PASS | WARN | FAIL | UNKNOWN",
  "reason":  "grids match; per-kidney registration declared; residual centroid shift under 1 voxel",
}
```

VERDICT

PASS	grids match (or transforms supplied) and registration scope is per_kidney or per_kidney_per_slice
WARN	a single global rigid transform was used for both kidneys — R8.1 says they must be registered separately
WARN	residual centroid shift > 1 in-plane voxel on either kidney
FAIL	grid/affine mismatch with no transform supplied — every ROI statistic would be meaningless
UNKNOWN	no affines available, or no registration provenance and no masks to run the residual test on

R8.1 (13% abstentions, 100% agreement): "retrospective image registration is highly recommended", with the body note "the kidneys should be registered separately if using rigid/affine transformations as they move independently." Corroborated by bones2019 (per-kidney translation-only Euler, Elastix), bones2022 (per-subject/per-kidney/per-slice, PCA groupwise metric) and olsen2025 (per-repetition-and-kidney rigid affine). This has no brain analogue at all. Grid/affine equality is a definition; the 1-voxel residual WARN is UNCALIBRATED — the same informal line as cox2017's visual "exclude > 1 voxel movement" rule.

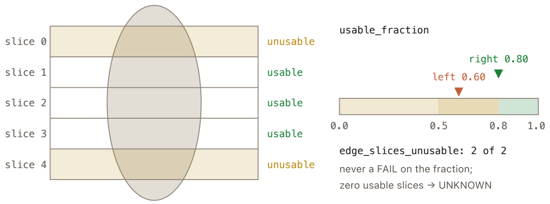
Needs affines and shapes for ΔM, M0 and the masks; registration provenance for the scope test; masks plus the 4D series for the optional residual test. When scope is absent but grids match, the check returns PASS on geometry with registration_scope: None stated in the reason — a reader is never told per-kidney registration happened when nobody said it did. Caveat: heavy background suppression can break ASL-driven registration to M0 — 54% success using the BGS ASL images versus 100% using separately-acquired fat images (bones2019, n=9), so read this alongside K8.3 rather than as a proxy for it.

K4.3 Slice coverage / usable-slice fraction

How many acquired slices actually survived to contribute to the reported value?

Catches a perfusion map assembled from fewer slices than were acquired, and the specific pattern where the losses are all at the ends of the stack — which tells the user their slab was positioned too tightly around the kidney.

The kidney runs past the slab, so the ends of the stack are lost first.



olsen2025: only 48.3% of 60 single-kidney scans retained all five slices — slice loss is normal, so it is reported, not failed on.

1. Resolve the slice axis from the affine — the array axis whose direction cosine is most nearly parallel to the slice normal. Never assume axis 2.
2. For each slice index along that axis and each kidney: count mask voxels, and count mask voxels where the perfusion value is finite and non-zero.
3. A slice counts if it has at least min_mask_voxels_per_slice mask voxels, and is usable if at least 90% of those voxels carry finite non-zero data.
4. usable_fraction = n_slices_usable / n_slices_with_mask, per kidney.
5. Report edge slices separately: how many unusable slices are first or last in the masked stack. Edge failure is a known systematic artefact, and "2 of 2 losses were edge slices" tells a user something actionable. Return N/A for 2D single-slice readout — coverage is 1/1 by construction and grading it would be theatre.

> Needs a perfusion map or mean ΔM, kidney masks, the affine, and the readout type from K5.2. Single-slice readout → N/A; no masks → UNKNOWN.

INPUT

```
{
  "rbf_map":      "NIfTI 3D float",     # or the mean delta_m
  "kidney_masks": {"left": "NIfTI 3D bool", "right": "NIfTI 3D bool"},
  "affine":       "4x4 float",          # to resolve which array axis is the slice axis
  "readout":      "2D multislice",      # from K5.2; "2D single-slice" makes this N/A
  "min_mask_voxels_per_slice": 20,
}
```

OUTPUT

```
{
  "metric": {
    "slice_axis": 2,
    "left":  {"n_slices_with_mask": 5, "n_slices_usable": 3, "usable_fraction": 0.60,
              "unusable_slice_indices": [0, 4], "edge_slices_unusable": 2},
    "right": {"n_slices_with_mask": 5, "n_slices_usable": 4, "usable_fraction": 0.80,
              "unusable_slice_indices": [4], "edge_slices_unusable": 1},
  },
  "verdict": "PASS | WARN | UNKNOWN | N/A",
  "reason":  "left kidney: 3 of 5 slices usable (0.60); both unusable slices are stack edges",
}
```

VERDICT

PASS	usable fraction ≥ 0.8 on both kidneys
WARN	usable fraction in 0.5–0.8 on either kidney
WARN	usable fraction < 0.5 on either kidney — reported prominently, still a WARN because the cut-points are uncalibrated
FAIL	never on the fraction. Fewer than one usable slice per kidney is reported as UNKNOWN, since no ROI statistic exists to grade
N/A	readout is 2D single-slice (R6.1, the consensus default)
UNKNOWN	no mask, no perfusion data, or zero usable slices

> 0.8 / 0.5 cut-points are UNCALIBRATED throughout — no paper states a usable-slice fraction criterion. The loss is heavily documented even though the threshold is not: in a 5-slice FAIR protocol only 48.3% of 60 single-kidney scans retained all five slices, the full distribution being 1.7% / 8.3% / 16.7% / 25.0% / 48.3% for one to five usable slices (olsen2025) — dropped by visual assessment, not by any metric. Edge slices fail systematically: "maps with significant noise were excluded (mostly generated from slices at the very beginning or end of the imaging stack, mainly due to partial volume effect with the nearby structures) … The middle slice … was the most representative one" (radovic2022). Coverage is protocol-conditional: R6.1 (10% abst / 95%) makes 2D single-slice the default, so whole-kidney coverage must never be assumed. The 90%-finite rule and the 20-voxel floor are construction choices, exposed as config.

Module K5 — Schema, data type & control/label ordering

Stream A. The module that works out what the scan actually is, and therefore which of the other checks are allowed to run.

K5 never grades image quality — it decides which of the other checks may run at all.

INPUT	K5 CHECK	WHAT IT GATES
sidecar · vendor header affine · K5.2 inference	K5.1 metadata completeness 12 gating items enforced, 25 reported PUBLISHED — Nery Table 4, 81–100%	K8.3 · K8.4 · K7.2 degrade to UNKNOWN, never FAIL
file names · shapes voxel sizes · affines	K5.2 data-type & geometry filename tokens + affine give the axes INFO only — it routes, never grades	K8.1 · K8.2 · K8.3 N/A, not a wrong verdict
4D series · kidney masks BS flag from K5.2	K5.3 control/label ordering per-pair sign count over the mask UNCALIBRATED bounds — 75% / 25%	the sign of every RBF value the only K5 check that FAILs

R10.1 (100%) makes every downstream report per kidney — so K5.2 must resolve laterality before Stream B can run.
R7.2 (5% abst / 80%) recommends background suppression, so K5.3 is often N/A on conformant renal data.

K5.1 Metadata completeness vs Nery Table 4

Is enough acquisition metadata present to run the checks that depend on it — above all Module K8?

It closes the upstream gap that would otherwise make eight downstream checks guess in silence.

enforce an item only if some other check reads it — report the rest

missing ⇒ the checks that consume it degrade to UNKNOWN. FAIL is not reachable.

resolved · missing

GATING ITEM (12)	GATES
labelling type	K5.2 · K8.1 · K8.2
inflow time / PLD(s)	K8.1 · K8.2 · K2.2
labelling duration	K8.2
number of averages	K8.2 · K7.2
background supp. + pulses	K5.3 · K6.2 · K8.3 · K8.4
repetition time	K8.2 · K6.3
slice thickness	K8.3
in-plane res. (affine)	K8.3
image orientation (affine)	K8.3
readout sequence type	K8.3
field strength	K8.4 · K6.3
organ + laterality	every per-kidney check

SOURCE PRIORITY — FIRST WINS

1 sidecar JSON
2 vendor header
3 affine-derived
4 K5.2 inference

checks_degraded
K8.3.readout
K8.4.alpha_bgs
K7.2.surviving_pairs

table4_reported_present 17 / 25 — counted for the report, never graded.

BIDS has no kidney: kidney, renal and abdom occur zero times in the spec; brain occurs 64 times.

1. Merge the sources in priority order: sidecar, then vendor header, then affine-derived, then K5.2's inference — and record which source each item came from.
2. Test the 12 gating items for presence and basic unit sanity: a PLD of 1400 is ms, a PLD of 1.4 is s; normalise both to seconds and record the assumed unit.
3. Map each missing item to the checks it disables via a static dependency table, and emit that as checks_degraded so the consequence sits in one place instead of eight scattered UNKNOWNs.
4. Count the full Table-4 set for the report, but do not grade on it.
5. Never FAIL, and emit organ/laterality under a clearly-named non-standard extension namespace so nothing is presented as BIDS-compliant that is not.

INPUT

```
{
  "sidecar":  {...parsed JSON...} | None,   # the CONTENTS, already loaded — not a path
  "header":   {...vendor DICOM-derived fields...} | None,
  "affine":   "4x4 float | None",           # supplies orientation and in-plane resolution
  "detected": {...},                        # K5.2's inference dict, used where metadata is absent
}
```

OUTPUT

```
{
  "metric": {
    "gating_present": ["ArterialSpinLabelingType", "PostLabelingDelay", "LabelingDuration",
                       "MagneticFieldStrength", "AcquisitionVoxelSize", "orientation"],
    "gating_missing": ["BackgroundSuppressionNumberPulses", "TotalAcquiredPairs",
                       "PulseSequenceType"],
    "table4_reported_present": 17, "table4_total": 25,
    "nonstandard_extension_used": {"organ": "kidney", "laterality": "bilateral"},
    "checks_degraded": ["K8.3.readout", "K8.4.alpha_bgs", "K7.2.surviving_pairs"],
  },
  ... 3 more lines
}
```

VERDICT

PASS	all 12 gating items resolvable from sidecar, header or affine
WARN	one or more gating items missing — they are listed, with the checks they degrade
FAIL	not reachable. No standard mandates renal ASL metadata, so its absence cannot fail a scan
UNKNOWN	no sidecar, no header and no affine — nothing to inspect at all

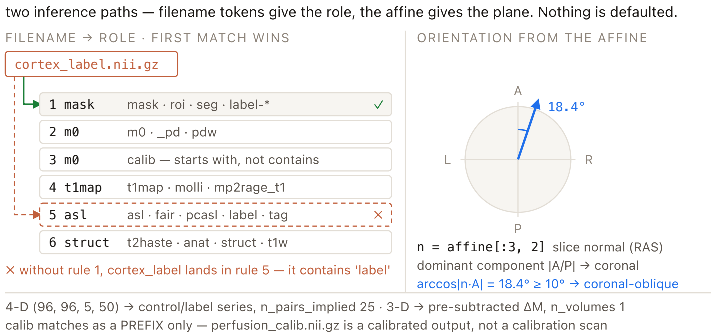
Nery 2020 Table 4, "Minimum set of parameters to be reported in ASL studies", agreed 81–100% — PUBLISHED, and a presence test, so it needs no calibration

BIDS does not cover the kidney — kidney, renal and abdom occur zero times in BIDS v1.11.1 against 64 occurrences of brain, and there is no registered BEP for body imaging. Two traps are encoded in the reader: BIDS SliceThickness is the microscopy field in micrometres, so Nery's slice thickness maps to AcquisitionVoxelSize; and LabelingSlabThickness says "for non-selective FAIR a zero is entered" while the schema sets exclusiveMinimum: 0 — FAIR is the dominant renal PASL variant, so renal data hits this. The reader tolerates 0 and the finding is filed upstream.

K5.2 Data-type & geometry detection · routing, emits INFO

What kind of renal acquisition is this, so that the right checks run and the wrong ones are marked N/A rather than failed?

It catches nothing on its own — it prevents mis-application: a 20-pair minimum enforced on a 3D acquisition, a QUIPSS II requirement applied to PCASL, or a control/label swap test run on a pre-subtracted map.



1. Give every file a role from its name, rules applied in order, first match wins: mask, then m0, then calib as a prefix only, then t1map, then asl, then struct.
2. Classify the labelling scheme from tokens only, never by default — R3.1 endorses both PCASL and FAIR, so with no token and no sidecar the value stays "unknown" and K8.1/K8.2 both return UNKNOWN.
3. Read the shapes: a 3-D ASL file is a pre-subtracted deltaM; a 4-D file is a control/label series with n_pairs_implied = N // 2; any m0-role file makes M0 "separate".
4. Resolve orientation from the affine, not the array order: the dominant RAS component of the slice normal names the base plane, arccos of that component gives the obliquity, and 10 degrees or more adds the -oblique suffix.
5. Background suppression is tri-state and never confidently false; emit INFO and let each downstream check read the axes and decide its own N/A.

INPUT

```
{
  "files": [{"name": "sub-01_asl.nii.gz", "shape": (96, 96, 5, 50),
             "voxel_mm": (3.0, 3.0, 5.0), "affine": "4x4 float"}, ...],
  "context": "Renal_FAIR_3T_native",      # the folder name
  "sidecar": {...} | None,                # used when present; inference is the fallback
}
```

OUTPUT

```
{
  "metric": {
    "labelling":      "FAIR",        # | "PCASL" | "VSASL" | "unknown"   (never assumed)
    "timing":         "single-TI",   # | "multi-TI" | "multi-PLD" | "unknown"
    "readout":        "2D single-slice",  # | "2D multislice" | "3D" | "unknown"
    "orientation":    "coronal-oblique",  # | "coronal" | "sagittal-oblique" | "axial"
    "obliquity_deg":  18.4,
    "structure":      "control/label series (50 volumes)",   # | "pre-subtracted deltaM"
    "n_volumes":      50, "n_pairs_implied": 25,
    "m0":             "separate",    # | "absent"
  ... 7 more lines
}
```

VERDICT

INFO	always, when files were supplied — it classifies and routes, it never grades
UNKNOWN	no files were passed in at all

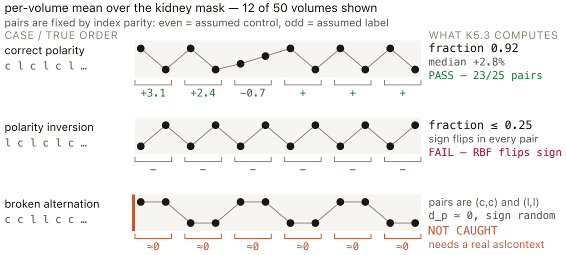
R3.1 both PCASL and FAIR endorsed (6% abst / 93%), R3.2 single time point (86%), R2.1 1.5 T and 3 T both adequate (87%), R6.7 coronal-oblique (6% abst / 93%) — the 10° obliquity cut is UNCALIBRATED and changes only a label string, never a verdict

Two orderings are load-bearing. mask must be matched before everything, because cortex_label.nii.gz contains "label" and would otherwise be filed as an ASL volume. And calib matches as a prefix only, because perfusion_calib is a calibrated output, not a calibration scan. Renal data in the wild is at least as metadata-poor as the three brain vendor cases, and the routing consequences here are sharper because all of Stream B is per-kidney, which requires laterality to be resolved.

K5.3 Control vs label ordering

Is the control/label polarity the right way round — is the control volume brighter than the label volume, consistently, across pairs?

It catches an inverted control/label polarity — the catastrophic case where every RBF value flips negative, the same failure K2.3's >20%-negative FAIL sees from the other side.



graded on control_brighter_fraction only — ≥ 0.75 PASS · 25–75% WARN · ≤ 0.25 FAIL

1. Gate first: background suppression ON → N/A, pre-subtracted ΔM → N/A, fewer than two volumes → N/A.
2. Split by index parity — even assumed control, odd assumed label — and write that assumption into the metric so no reader mistakes it for a measured fact; a supplied aslcontext overrides it.
3. Compute per pair, not pooled: c_p and l_p are nanmeans over the kidney mask, and $d_p = 100 \cdot (c_p - l_p) / (0.5 \cdot (c_p + l_p))$.
4. Grade control_brighter_fraction = count($d_p > 0$) / n_pairs, and report the median d_p as the motion-robust central estimate.
5. Pool both kidneys for this one test only — polarity is a property of the sequence, not of an organ, and pooling doubles the voxel count behind each pair mean.

INPUT

```
{
  "asl_4d":        "NIfTI 4D float",     # even index ASSUMED control, odd ASSUMED label
  "kidney_masks":  {"left": "NIfTI 3D bool", "right": "NIfTI 3D bool"},
  "background_suppression": False,        # True | False | None; True makes the check N/A
  "structure":     "control/label series (50 volumes)",    # from K5.2
  "aslcontext":    None,                  # honoured if supplied; nothing in v1 produces one
}
```

OUTPUT

```
{
  "metric": {
    "n_pairs": 25,
    "control_brighter_fraction": 0.92,          # the graded statistic
    "median_rel_diff_pct": 2.8,                 # median over pairs of 100*(c−l)/mean(c,l)
    "per_pair_rel_diff_pct": [3.1, 2.4, −0.7, ...],
    "assumption": "even=control (no aslcontext supplied); BS assumed off",
    "roi": "kidney masks, both sides pooled for this test only",
  },
  "verdict": "PASS | WARN | FAIL | UNKNOWN | N/A",
  ... 2 more lines
}
```

VERDICT

PASS	control brighter in ≥ 75% of pairs
WARN	control brighter in 25–75% of pairs — inconsistent sign, motion is dominating the label difference; the subtraction may still be correct but this scan cannot confirm it
FAIL	control brighter in ≤ 25% of pairs — the label slab is consistently the brighter one, so the subtraction runs backwards and every RBF value flips sign
N/A	background suppression ON, pre-subtracted ΔM, or fewer than two volumes
UNKNOWN	no 4D series, no kidney mask, or every pair mean is non-finite

Control brighter than label is physics, not a tuned number, and that is what the FAIL rests on; the 75% / 25% consistency bounds are UNCALIBRATED — no published renal swap threshold exists. R7.2 (5% abst / 80%) recommends background suppression for the pairs, so this check is frequently N/A on conformant renal data.

Stated limitation: it does NOT catch a broken alternation such as c,c,l,l — that averages to the same fraction and is invisible to an intensity comparison; only a real aslcontext catches it. The reason the brain's pooled two-slab comparison was replaced by per-pair sign counting: renal acquisitions are free breathing by consensus (R7.4, 0% abst / 76%), and volume-to-volume swings from the kidney moving through the slice can exceed a 3% label difference. The uncalibrated bounds can only make the FAIL harder to reach than the brain equivalent, never easier — an uncalibrated number that can only suppress a FAIL is a safe one.

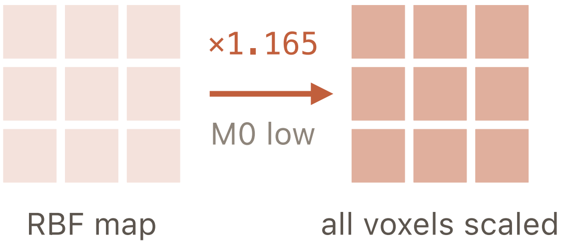
Module K6 — M0 calibration

The one module that transfers from brain unchanged: it guards the denominator that every RBF voxel is divided by.

M0 is the denominator of every voxel — so an M0 error scales the whole map, uniformly

for every voxel x

$$\frac{\Delta M(x)}{M0} \times \text{const} =$$



what Module K2 / K3 sees afterwards

cortex : medulla	unchanged
left : right	unchanged
negative fraction	unchanged

invisible to Stream B

K6.1
M0 present at all
R9.1 · 94% agreement

K6.2
no BS, no labelling
nery2020 p.13, verbatim

K6.3
TR ≥ 5 s, or corrected
1 / (1 − e^(−TR/T1))

K6.1 — M0 present

Before we grade a single RBF number, we ask whether a calibration image exists at all — the image the consensus calls mandatory.

It stops a dataset whose RBF values were produced with no recorded calibration reference — an error that scales every voxel by an unknown constant and that no Stream-B check can see.

no calibration image ⇒ no denominator ⇒ the numbers being graded are not RBF

`m0_type = "separate"` or `"included"`	$\frac{\Delta M}{M0}$	denominator recorded	PASS
`m0_type = "absent"` `rbf_supplied = True`	$\frac{\Delta M}{?}$	no denominator exists	FAIL
`m0_type = "absent"` `delta_m_only = True`	ΔM only	no quantification claimed	WARN

`m0_type = None → UNKNOWN (no detection ran)`

1. Read m0_type from K5.2's detection: a file whose name contains m0 / _pd, or starts with calib. No voxel data is touched.
2. separate or included → PASS.
3. absent AND a quantified RBF map is being graded → FAIL: there is no denominator, so the numbers rest on a calibration nobody recorded.
4. absent AND only ΔM was supplied → WARN: no quantification is claimed, so it limits what can be checked rather than invalidating a result.
5. None — no detection ran — → UNKNOWN.

INPUT

```
{
  "m0_type":       "separate",   # | "included" | "absent" | None – from K5.2
  "rbf_supplied":  True,         # is a quantified RBF map being graded?
  "delta_m_only":  False,        # only a perfusion-weighted difference image was supplied
}
```

OUTPUT

```
{
  "metric": {"m0_type": "separate", "quantification_claimed": True},
  "verdict": "PASS | WARN | FAIL | UNKNOWN",
  "reason":  "M0 present (separate file)",
}
```

VERDICT

PASS	m0_type is separate or included
FAIL	m0_type is absent AND a quantified RBF map is being graded
WARN	m0_type is absent and only a ΔM / perfusion-weighted image was supplied
UNKNOWN	m0_type was never determined

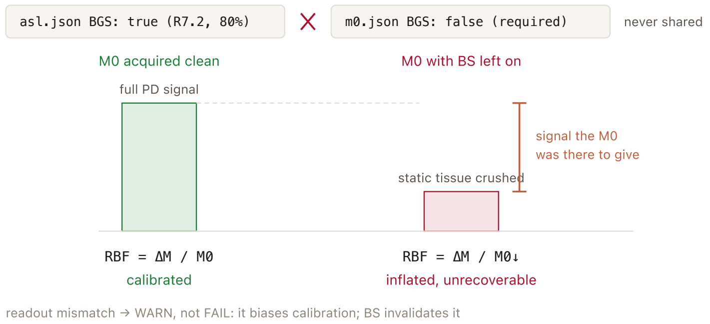
R9.1 — 0% abstentions, 94% agreement: "M0 acquisition is mandatory". Stronger than the brain White Paper (which only recommends a separate PD image) and stronger than BIDS (which permits M0Type: "Absent"). Presence test is binary file detection — nothing to tune.

> Why is a FAIL allowed here when almost nothing else in this design FAILs? Agreement is ≥ 90% and the deviation invalidates rather than degrades: without a calibration image, the quantity being graded is not the quantity the protocol defines. This is one of the very few places a renal check is legitimately stricter than its brain counterpart, and it has a published warrant.

K6.2 — M0 acquired without labelling and without background suppression

Then we ask whether that calibration image was acquired clean — because background suppression destroys exactly the signal the M0 is there to provide.

It catches an M0 mistakenly acquired with the ASL protocol's background suppression left on — a setup that quietly inflates every RBF value by a factor nobody can recover after the fact.



1. Read BackgroundSuppression and the labelling flag from the M0 scan's OWN metadata, never from the ASL pairs' metadata — the two volumes follow opposite rules.
2. Both false → PASS.
3. Either true → FAIL: the denominator is no longer proton-density weighted, so every RBF value is scaled wrong.
4. Compare m0_readout against asl_readout as a string/tuple match; a mismatch is a WARN, not a FAIL.
5. Either flag absent → UNKNOWN. No voxel data is touched by this check.

INPUT

```
{
  "m0_background_suppression": False,   # True | False | None – read from the M0's OWN metadata
  "m0_labelling_applied":      False,   # True | False | None
  "m0_readout":                "2D SE-EPI",
  "asl_readout":               "2D SE-EPI",
}
```

OUTPUT

```
{
  "metric": {
    "m0_background_suppression": False,
    "m0_labelling_applied":      False,
    "readout_matches_asl":       True,
  },
  "verdict": "PASS | WARN | FAIL | UNKNOWN",
  "reason":  "M0 acquired without labelling or background suppression, readout matches ASL",
}
```

VERDICT

PASS	no labelling AND no background suppression on the M0, readout matched
WARN	flags clean but the M0 readout differs from the ASL readout
FAIL	background suppression was ON, or labelling was applied, during the M0 acquisition
UNKNOWN	the M0's own BS / labelling flags are not recorded

nery2020 p.13, verbatim: "The PD image should be acquired without labelling or BGS and using a similar readout and acquisition parameters, with the exception that a long TR should be used." Identical rule to the brain White Paper — it ports unchanged. R7.2 (5% abstentions, 80% agreement) recommends BGS for the ASL pairs, which is not a contradiction. Both flags are binary reads — nothing to tune.

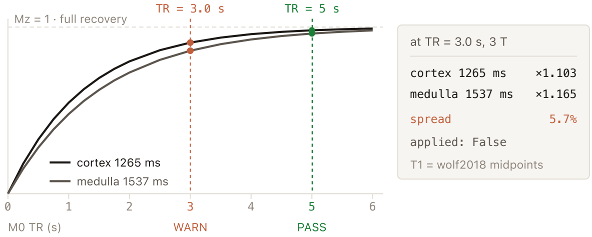
Why not reuse K5.2's background-suppression inference here? Because K5.2 infers BS from the ASL folder's text, and the M0 follows the opposite rule — borrowing that flag would assert exactly the wrong conclusion. Missing M0 flags stay UNKNOWN rather than being filled in.

K6.3 — M0 TR ≥ 5 s, or corrected

And finally, whether the M0 waited long enough for tissue to recover — and if it did not, how big the correction is and how uncertain that correction really is.

It catches a too-fast M0 acquisition, which gives an artificially low calibration denominator and therefore inflated RBF — flagged and quantified rather than silently corrected.

how much tissue signal the M0 was allowed to recover before it was read



f(T1) = 1 / (1 − exp(−TR / T1)) nery2020 p.13, verbatim FAIL: never

1. TR ≥ 5 s → PASS, correction_factor = 1.0, nothing further to do.
2. TR < 5 s → compute f(T1) = 1 / (1 − exp(−TR / T1)) in pure NumPy, twice: once with the cortex T1, once with the medulla T1 for that field strength.
3. Defaults are the midpoints of the published wolf2018 ranges: 3 T cortex 1265 ms, medulla 1537 ms; 1.5 T cortex 953 ms, medulla 1241 ms.
4. Report both factors and their spread and emit WARN — do not silently apply one; applied stays False.
5. If a quantitative t1_map is supplied, correct per compartment using ROI-mean T1, set applied: True. Missing TR → UNKNOWN.

INPUT

```
{
  "m0_tr_s":   3.0,          # seconds
  "field_T":   3.0,          # selects the default compartment T1s
  "t1_map":    None,         # NIfTI 3D float in ms; if present, correction is per-compartment
  "cortex_masks":  {...},    # only used when t1_map is supplied
  "medulla_masks": {...},
}
```

OUTPUT

```
{
  "metric": {
    "tr_seconds": 3.0,
    "correction_factor_cortex":  1.103,
    "correction_factor_medulla": 1.165,
    "t1_used_ms": {"cortex": 1265, "medulla": 1537, "source": "wolf2018 range midpoint"},
    "correction_spread_pct": 5.7,
    "applied": False,
  },
  "verdict": "PASS | WARN | UNKNOWN",
  ... 2 more lines
}
```

VERDICT

PASS	M0 TR ≥ 5 s
WARN	TR < 5 s — both correction factors and their spread are reported; the correction is not applied unless a T1 map was supplied
FAIL	never — the consensus states this as a correctable condition, not a disqualifying one
UNKNOWN	no M0 TR recorded

The 5 s rule and the exact formula are PUBLISHED and directly portable — nery2020 p.13 verbatim: "If this image is acquired without waiting for a sufficiently (> 5 s) long recovery time (TR), the SI_PD should be corrected for incomplete relaxation ... where T1,tissue is an estimate of the kidney T1." But the consensus gives no numeric kidney T1 anywhere, so the T1 defaults are UNCALIBRATED as a choice (ranges from wolf2018).

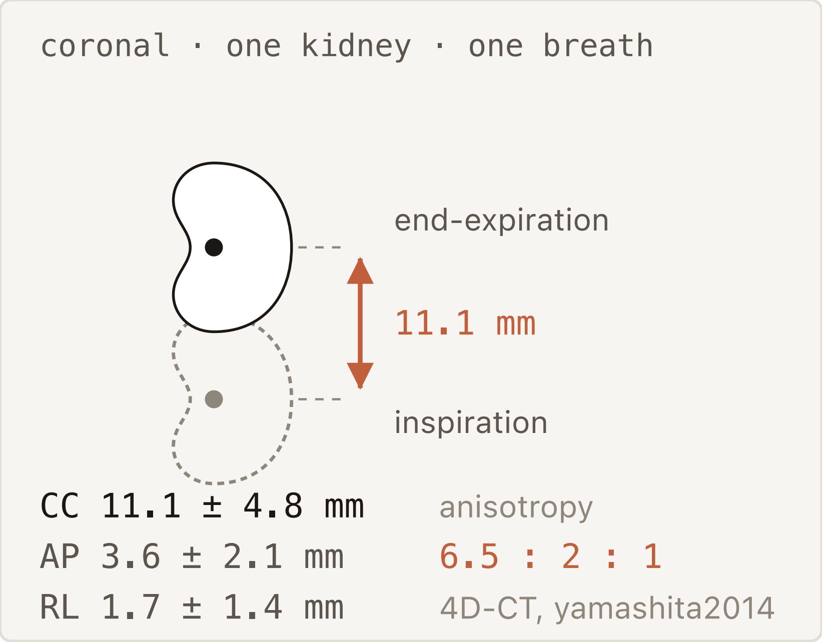
Why report two factors instead of picking one? Kidney has two tissue T1s ~400 ms apart, so the two corrections differ by 2.6–9.6% depending on TR — comparable to the technical effects K3.1 already documents. Hiding that inside a single silent correction would inject an error the same size as the ones this design spends pages warning about. A supplied T1 map is the only path where one measured number is used.

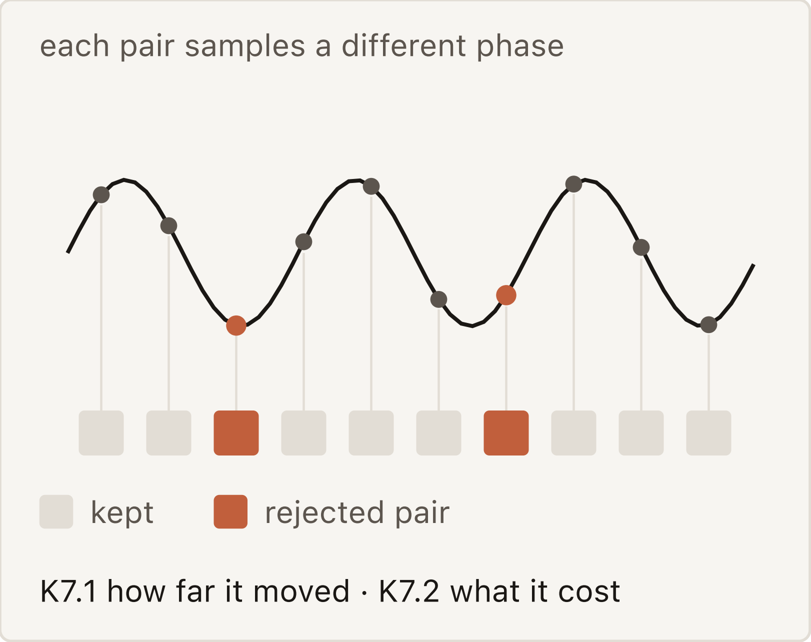
MODULE K7

K7

Module K7 — Respiratory motion & outlier rejection

The module with no brain equivalent: measure how far each kidney moved, then count what that motion actually cost.



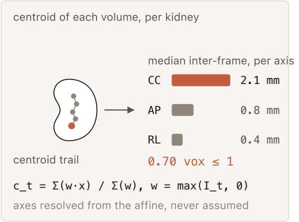


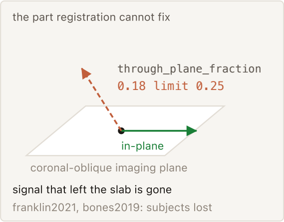
R8.1 registration 100% · R8.2 outlier rejection 100% · no method, no threshold stated

K7.1 · Per-kidney respiratory displacement

How far did each kidney move, along which axis, during the series?

It stops a map being read as clean when the kidney it came from swam several voxels — and separates in-plane motion, which registration fixes, from through-plane motion, which it does not.





1. Resolve CC / AP / RL from the affine's direction cosines — derive the axes, never assume array-axis order.
2. Per kidney, per volume, take the intensity-weighted centroid inside the mask: $c_t = \Sigma(w \cdot x)/\Sigma(w)$ with $w = \max(I_t, 0)$. Pure NumPy, no registration needed. If per-volume transforms are supplied, use their translations instead.
3. Project consecutive-volume displacements $c_t - c_{t-1}$ onto the CC, AP and RL unit vectors.
4. Report median and max per axis, in mm and in voxels — voxels is the decision-relevant unit, since 11 mm is catastrophic at 2 mm in-plane and tolerable at 8 mm slices.
5. Add through_plane_fraction, the share of displacement normal to the imaging plane. Never collapse to one scalar: an isotropic brain-FD-style sum would average away the CC axis, which carries ~6.5× the RL motion.

INPUT

```
{
  "delta_m_4d":    "NIfTI 4D float",       # or the raw control/label series
  "kidney_masks":  {"left": "NIfTI 3D bool", "right": "NIfTI 3D bool"},
  "voxel_size_mm": [3.0, 3.0, 5.0],
  "affine":        "4x4 float",             # to resolve which axis is cranio-caudal
  "transforms":    None,                    # optional per-kidney per-volume translations
}
```

OUTPUT

```
{
  "metric": {
    "left": {
      "median_interframe_cc_mm": 2.1, "max_cc_excursion_mm": 9.4,
      "median_interframe_ap_mm": 0.8, "median_interframe_rl_mm": 0.4,
      "median_interframe_cc_vox": 0.70,   # the units that actually matter
      "through_plane_fraction": 0.18,     # normal to the imaging plane
      "n_frames_used": 50,
    },
    "right": {...},
  ... 6 more lines
}
```

VERDICT

PASS	median inter-frame CC displacement ≤ 1 voxel on both kidneys AND through-plane fraction ≤ 0.25
WARN	median inter-frame CC displacement > 1 voxel on either kidney
WARN	through-plane fraction > 0.25 on either kidney — the component registration cannot correct
FAIL	not on this metric. Severity is reported here; the consequence — how many pairs it corrupted — is graded by K7.2
UNKNOWN	no 4D series, no mask, no affine, or fewer than 3 usable frames; a pre-subtracted single ΔM volume has nothing to move between frames

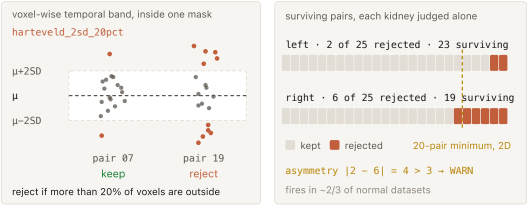
UNCALIBRATED — there is NO published mm or voxel threshold for failing a renal ASL scan on motion; the 1-voxel line comes from cox2017's visual pipeline rule "exclude > 1 voxel movement", which never states the voxel size, and the 0.25 through-plane line is an engineering default

Why not FD/DVARS? A measured claim: "framewise displacement" AND (kidney OR renal perfusion) AND MRI NOT brain returns 0 hits on Europe PMC. Structurally, renal motion is anisotropic, the two kidneys move independently so a single 6-DOF series is ill-defined, and brain FD converts rotations on a 50 mm head-radius sphere that is meaningless for an abdominal organ. Through-plane motion is a documented cause of total data loss: one subject of 6 excluded in franklin2021, one of 10 in bones2019.

K7.2 · Subtraction-outlier rate ⭐

How many control–label pairs were corrupted enough to reject — the direct operationalisation of R8.2, which recommends outlier rejection and specifies no way to do it.

It measures the actual consequence of respiratory motion — how much of the acquisition was destroyed — which K7.1 cannot give, since a large excursion that registration handled costs nothing and a small one at the wrong moment costs a pair.



1. Per kidney, build the voxel-wise temporal mean and SD of ΔM across repetitions, inside the mask: mu = nanmean(dM, axis=3), sd = nanstd(dM, axis=3, ddof=1).
2. For each pair p, compute the fraction of within-mask voxels with abs(dM[..., p] − mu) > k · sd.
3. Reject the pair if that fraction exceeds the rule's limit — the default harteveld_2sd_20pct rejects when more than 20% of kidney voxels deviate beyond ±2 SD.
4. Run each kidney independently — one kidney can be corrupted while the other is clean — and report n_rejected, rejected_fraction, surviving_pairs and the left-right asymmetry.
5. Gate the surviving-pair comparison on readout: the 20-pair minimum applies to 2D only, so on a 3D readout consensus_minimum_applicable is None.

INPUT

```
{
  "delta_m_4d":   "NIfTI 4D float",       # (X, Y, Z, n_pairs)
  "kidney_masks": {"left": "NIfTI 3D bool", "right": "NIfTI 3D bool"},
  "rule":         "harteveld_2sd_20pct",  # one of four published parameterisations
  "readout":      "2D single-slice",       # from K5.2; gates the surviving-pair comparison
  "n_pairs_acquired": 25,
}
```

OUTPUT

```
{
  "metric": {
    "rule": "harteveld_2sd_20pct",
    "left":  {"n_pairs": 25, "n_rejected": 2, "rejected_fraction": 0.08, "surviving_pairs": 23},
    "right": {"n_pairs": 25, "n_rejected": 6, "rejected_fraction": 0.24, "surviving_pairs": 19},
    "consensus_minimum_applicable": 20,   # None on 3D readouts
    "asymmetry": 4,                       # |n_rejected_left − n_rejected_right|
    "note": "the rule fires in ~2/3 of NORMAL healthy datasets – the count is the signal",
  },
  "verdict": "PASS | WARN | FAIL | UNKNOWN",
  ... 2 more lines
}
```

VERDICT

PASS	≤ 2 pairs rejected per kidney AND (on 2D readouts) surviving pairs ≥ 20
WARN	3 or more pairs rejected on either kidney
WARN	surviving pairs < 20 on a 2D readout — the consensus minimum is no longer met after rejection
WARN	rejection count differs by more than 3 between the two kidneys — one-sided corruption
FAIL	fewer than 2 pairs survive on either kidney — no averaging is possible at all
N/A	pre-subtracted ΔM, or fewer than 4 repetitions (the temporal SD is not estimable)
UNKNOWN	no 4D series or no mask

R8.2 outlier rejection, 0% abst / 100% — the strongest agreement in Table 2, and it states NO criterion; the four rules are IMPLEMENTATION (harteveld2020 default, bones2021, harteveld2022 paediatric 1.5 SD, garciaruiz2025 +RMSE) and the ≤ 2-rejected PASS line is UNCALIBRATED; the 20-pair minimum is R4.7 (10% abst / 89%) and R5.12 (14% abst / 83%)

> The rule firing is not itself a defect signal — Harteveld's rule fired in 18/27 FAIR and 19/27 pCASL healthy datasets, with a maximum of 2 excluded pairs per delay. Only the count matters, and that observed ceiling in known-good data is what the PASS line is anchored on. Falling below 20 surviving pairs is a WARN and not a FAIL for two stated reasons: at 83-89% it sits below the ≥ 90% agreement bar this design requires for a FAIL, and it is an acquisition-planning number being applied to a post-hoc surviving count — a deliberate extrapolation, and this design does not escalate on its own extrapolations.

K7.3 · Breathing strategy & gating efficiency

Was a consensus-conformant breathing strategy used, and if a respiratory trace exists, how much data survived gating?

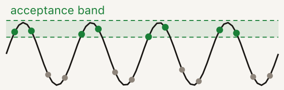
It catches a protocol that used breath-holds against a 94%-agreement recommendation, and — where a trace exists — a gating setup accepting far fewer windows than the acquisition assumed.

breathing strategy → verdict

free	PASS	R7.4 · 76%
triggered	PASS	R7.5 · 95%
paced	PASS	no number
breath_hold	WARN	R7.3 · 94%

no BIDS field → usually UNKNOWN

gating efficiency — reported, never graded



gating_efficiency_pct =
100 · n_accepted / n_total_windows

normal: 50 ± 13% coronal, 41 ± 9% sagittal

a 40–50% floor would fail normal scans

1. Read breathing_strategy from metadata. It has no BIDS field, so in practice it arrives via the non-standard extension namespace defined in K5.1, or not at all.
2. free or triggered → PASS; paced → PASS with the strategy named in the reason, since no numbered statement endorses or discourages it; breath_hold → WARN.
3. If a respiratory trace and trigger times are supplied, compute 100 · n_accepted_windows / n_total_windows, accepting a window when its trigger timestamp falls inside the acceptance band of the trace.
4. Report that efficiency as INFO inside the metric and emit no verdict on the number at all.

INPUT

```
{
  "breathing_strategy": "free",   # "free" | "triggered" | "paced" | "breath_hold" | None
  "physio_trace":       None,     # optional: {"respiratory": np.ndarray, "sampling_hz": 100.0}
  "trigger_times_s":    None,     # optional: acquisition timestamps, for gating efficiency
  "acquisition_time_s": 222.0,
}
```

OUTPUT

```
{
  "metric": {
    "breathing_strategy": "free",
    "conforms_to_R7_4":   True,
    "gating_efficiency_pct": None,   # populated only when a trace + trigger times are supplied
    "n_accepted_windows": None, "n_total_windows": None,
    "trace_present": False,
  },
  "verdict": "PASS | WARN | UNKNOWN",
  "reason":  "free breathing – conforms to R7.4 (76% agreement)",
}
```

VERDICT

PASS	free breathing, or free breathing with respiratory triggering
WARN	breath-hold — R7.3 (0% abst, 94%): "Breath-hold scans are not recommended for clinical renal ASL"
FAIL	never. R7.4 sits at 76%, one point above the consensus bar, and 94% for R7.3 supports a firm flag but not a rejection: breath-held renal ASL is discouraged practice, not invalid data
INFO	gating efficiency reported when a trace is present — ungraded
UNKNOWN	breathing strategy not recorded — and it has no BIDS field, so this is common

R7.3 breath-hold not recommended (0% abst, 94%) drives the WARN; R7.4 free breathing preferred (0% abst, 76%); R7.5 respiratory triggering (5% abst, 95%); gating efficiency is UNCALIBRATED and never graded

Gating efficiency is deliberately never thresholded: reference values are 50 ± 13% (coronal navigator) and 41 ± 9% (sagittal) in 10 healthy volunteers, so a "below 40-50% indicates poor compliance" rule would fail roughly half of normal scans — the cohort-mean-as-threshold error, named here so it is not re-introduced later. And a trace is never sufficient on its own: bones2020 found no one-to-one correlation between bellows signal and spurious labelling — "Some repetitions had tissue labeling without the bellows indicating motion" — so the image-derived K7.1 and K7.2 are required regardless.

Module K8 — Protocol conformance vs the renal consensus

Four straight metadata range tests against Nery et al. 2020 — the only module in the kidney design where every threshold is a numbered consensus statement with an agreement percentage attached.

the agreement percentage sets the MAXIMUM severity a deviation may reach

75 - 89% agreement

below the bar this design requires for a FAIL

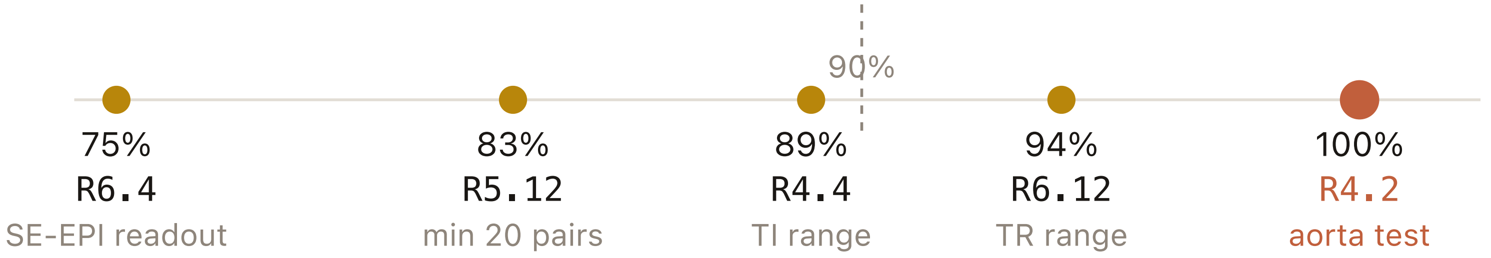
ceiling: WARN

≥ 90% and the deviation INVALIDATES

FAIL - R4.2 only, once in K8

≥ 90% but it only DEGRADES

WARN - bias reported, not rejected

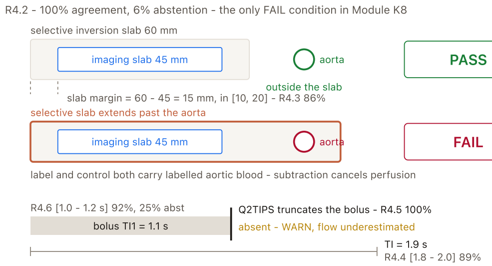


- abstention ≥ 25% - capped at WARN, the abstention stated in the reason
- abstention ≥ 40% (R5.10 only) - INFO, excluded from grading entirely

K8.1 — FAIR / PASL timing conformance

For a pulsed renal acquisition: do the labelling timings and the selective-slab geometry match the consensus recommendations — and does the selective slab keep the aorta out?

It catches a pulsed renal scan whose bolus is not width-controlled, whose TI sits outside the range the consensus timings assume, and — the one that can actually void the measurement — a selective slab that labels the aorta.



1. Gate on labelling: not PASL/FAIR → N/A; labelling unknown → UNKNOWN. Never guess — R3.1 endorses both schemes, so a default would be a coin flip.
2. Normalise units: accept ms or s for TI and TI1, convert to seconds, record the assumed unit.
3. slab_margin_mm = selective_slab_thickness_mm – imaging_slab_thickness_mm, tested against 10–20 mm.
4. Test each parameter against its recommended range or value; a parameter with no value recorded goes into n_unknown and contributes nothing to the verdict.
5. Apply the module escalation rule per parameter, take the worst outcome, name the specific statement in the reason — and if bolus saturation is absent, WARN with the bias direction spelled out.

INPUT

```
{
  "labelling":                    "FAIR",     # from K5.2; anything else makes this N/A
  "inversion_time_s":             1.9,        # TI
  "bolus_duration_s":             1.1,        # TI1
  "bolus_saturation":             "Q2TIPS",   # "QUIPSS II" | "Q2TIPS" | None
  "selective_slab_thickness_mm": 60.0,
  "imaging_slab_thickness_mm":   45.0,
  "selective_slab_excludes_aorta": True,      # True | False | None
  "inversion_pulse_type":         "FOCI",
}
```

OUTPUT

```
{
  "metric": {
    "checked": {
      "inversion_time_s":  {"value": 1.9, "recommended": [1.8, 2.0], "conforms": True,
                            "statement": "R4.4", "agreement": 89, "abstention": 10},
      "bolus_saturation":  {"value": "Q2TIPS", "recommended": "QUIPSS II or Q2TIPS",
                            "conforms": True, "statement": "R4.5", "agreement": 100,
                            "abstention": 25},
      "slab_excludes_aorta": {"value": True, "recommended": True, "conforms": True,
                            "statement": "R4.2", "agreement": 100, "abstention": 6},
  ... 9 more lines
}
```

VERDICT

PASS	every checked parameter conforms
WARN	any conformance failure other than the aorta test, including a missing QUIPSS II / Q2TIPS
FAIL	selective_slab_excludes_aorta is False — the selective inversion labels the aorta itself, so the FAIR subtraction cancels the very signal it is meant to isolate
N/A	labelling is PCASL or VSASL
UNKNOWN	labelling unknown, or no PASL timing fields recorded at all

All PUBLISHED, Nery 2020 Table 2 — R4.2 100% agreement / 6% abstention (the only FAIL in K8); R4.4 89%, R4.6 92% (25% abst), R4.5 100% (25% abst), R4.3 86%, R4.1 92%.

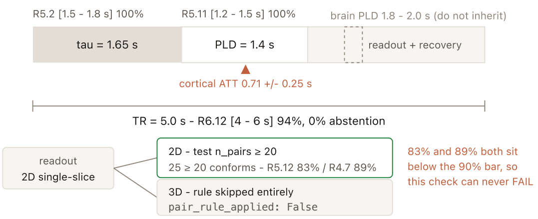
R4.5 has 100% agreement and still only WARNs — a stated decision. 25% of the panel abstained, and the paper itself concedes that replacing TI1 with TI "will underestimate flow by a transit time-dependent factor". A hard FAIL would reject a large slice of the published renal literature, so the check WARNs and puts that bias direction in the reason. This is softer than the brain White Paper.

K8.2 — PCASL timing & averaging conformance

For a pseudo-continuous renal acquisition: are the labelling duration, post-labelling delay, TR and pair count the renal numbers — and not the brain ones?

It catches a renal PCASL protocol built with brain constants — the single most likely misconfiguration for a site moving from neuro to body ASL — and an averaging count too low for the readout in use.

one PCASL pair, drawn to scale in time - every block has a consensus range



1. Gate on labelling: not PCASL → N/A; unknown → UNKNOWN.
2. Normalise ms/s and test τ, PLD and TR against their consensus ranges.
3. Gate the 20-pair rule on readout: if the readout is 3D, set pair_rule_applied: False and skip it entirely.
4. On 2D readouts, test n_pairs_acquired >= 20. Report n_pairs_surviving from K7.2 alongside, but do not grade it here — K7.2 owns the rejection accounting.
5. Apply the escalation rule per parameter; worst outcome wins; name the statement in the reason.

INPUT

```
{
  "labelling":            "PCASL",    # from K5.2; anything else makes this N/A
  "labeling_duration_s":  1.65,       # tau
  "post_labeling_delay_s": 1.4,
  "repetition_time_s":    5.0,        # labelling + readout
  "n_pairs_acquired":     25,
  "n_pairs_surviving":    23,         # from K7.2, when available
  "readout":              "2D single-slice",   # gates the 20-pair rule
}
```

OUTPUT

```
{
  "metric": {
    "checked": {
      "labeling_duration_s":   {"value": 1.65, "recommended": [1.5, 1.8], "conforms": True,
                                 "statement": "R5.2",  "agreement": 100, "abstention": 10},
      "post_labeling_delay_s": {"value": 1.4,  "recommended": [1.2, 1.5], "conforms": True,
                                 "statement": "R5.11", "agreement": 100, "abstention": 19},
      "repetition_time_s":     {"value": 5.0,  "recommended": [4.0, 6.0], "conforms": True,
                                 "statement": "R6.12", "agreement": 94,  "abstention": 0},
      "n_pairs_acquired":      {"value": 25,   "recommended": ">= 20",    "conforms": True,
  ... 9 more lines
}
```

VERDICT

PASS	every checked parameter conforms
WARN	τ, PLD or TR outside its consensus range, or (2D only) fewer than 20 pairs acquired
FAIL	never in this check — no statement here combines ≥ 90% agreement with an invalidating deviation
N/A	labelling is PASL/FAIR or VSASL
UNKNOWN	labelling unknown, or no PCASL timing fields recorded

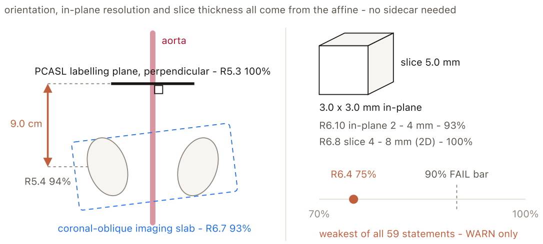
All PUBLISHED, Nery 2020 Table 2 (statements quoted from p.7) — R5.2 100% (10% abst), R5.11 100% (19%), R6.12 94% (0%); the pair rule R5.12 83% (14%) / R4.7 89% (10%), both below the 90% bar.

Two things must not be inherited from brain. PLD 1.2–1.5 s is shorter than the brain White Paper's 1.8–2.0 s and only marginally longer than the measured pCASL cortical ATT of 0.71 ± 0.25 s. And the 20-pair rule is scoped by the Discussion to "when using the recommended 2D readout" — a 3D-GRASE paediatric allograft study used 6 pairs because 20 would have taken 12 min 6 s instead of 3 min 42 s.

K8.3 — Geometry & readout conformance

Was the acquisition geometry — orientation, resolution, slice thickness, readout scheme, suppression options — the geometry the consensus recommends?

It catches a renal protocol assembled with brain-ASL geometry habits: a thin-slice 3D whole-organ acquisition where the consensus asks for a thick 2D single slice, an axial orientation that puts respiratory motion through-plane, or a PCASL labelling plane placed by brain convention rather than 8–10 cm above the kidneys.



1. Take orientation, in-plane resolution and slice thickness from the affine where metadata is absent — record geometry_source per item.
2. Test orientation against coronal-oblique (R6.7); for transplants invert the expectation — the allograft sits in the iliac fossa, so a non-coronal orientation is reported as conforming-by-context.
3. Test readout (R6.1/R6.3), pulse sequence (R6.4/R6.5), slice thickness 4–8 mm for 2D or 3–6 mm for 3D, in-plane 2–4 mm, acceleration ≤ 2, and the fat- and background-suppression flags.
4. Gate the labelling-plane rows on labelling == "PCASL"; they are skipped for FAIR.
5. Apply the escalation rule per parameter, worst outcome wins; metadata-only items (fat suppression, bandwidth, triggering, acceleration) are counted in n_unknown, never failed.

INPUT

```
{
  "orientation":        "coronal-oblique",   # from K5.2's affine analysis
  "readout":            "2D single-slice",
  "pulse_sequence_type": "SE-EPI",
  "in_plane_mm":        [3.0, 3.0],
  "slice_thickness_mm": 5.0,
  "parallel_imaging_factor": 2,
  "fat_suppression":    True,
  "background_suppression": True,
  "labelling":          "PCASL",             # gates the labelling-plane rows
  ... 4 more lines
}
```

OUTPUT

```
{
  "metric": {
    "checked": {
      "orientation":       {"value": "coronal-oblique", "recommended": "coronal oblique",
                            "conforms": True, "statement": "R6.7",  "agreement": 93, "abstention": 6},
      "pulse_sequence":    {"value": "SE-EPI", "recommended": "spin-echo EPI",
                            "conforms": True, "statement": "R6.4",  "agreement": 75, "abstention": 5},
      "slice_thickness_mm": {"value": 5.0, "recommended": [4, 8], "conforms": True,
                            "statement": "R6.8",  "agreement": 100, "abstention": 19},
      "in_plane_mm":       {"value": [3.0, 3.0], "recommended": [2, 4], "conforms": True,
  ... 14 more lines
}
```

VERDICT

PASS	every checked geometry parameter conforms
WARN	any deviation, naming the statement and its agreement in the reason
FAIL	never — nothing here invalidates a measurement; a thicker slice or coarser matrix produces worse renal ASL, not non-renal-ASL
UNKNOWN	neither metadata nor an affine — nothing to test against

All PUBLISHED, Nery 2020 Table 2 — R6.7 93%, R6.1 95%, R6.4 75% (5% abst, the lowest agreement of all 59 statements), R6.8 100%, R6.10 93%, R6.11 100%, R7.6 90%, R7.2 80%, R5.3 100% / R5.4 94%.

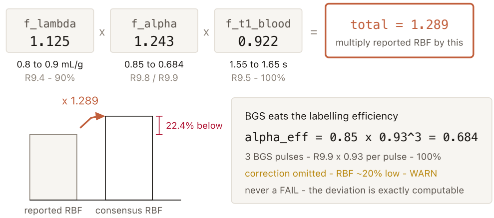
R6.4 verbatim (p.7): "Spin-echo EPI is the preferred readout for 2D single-slice acquisitions" — 75% agreement, sitting exactly on the inclusive consensus bar, so a check on it WARNs and can never do more. Note also that geometry is checkable from the affine with no sidecar at all, which makes K8.3 the most robust member of the module on real metadata-poor renal data.

K8.4 — Quantification-constant conformance

If someone hands me a quantified RBF map: were the constants used to turn ΔM into RBF the consensus constants — and if not, by exactly what factor is the reported number off?

It catches a reported RBF map that is correct arithmetic on the wrong constants — the failure that is completely invisible to every Stream-B check, because a uniformly rescaled map has the same cortex:medulla ratio, the same left-right asymmetry, the same negative fraction and the same distribution shape as a correct one.

the toolbox does not re-quantify - it reports the exact multiplicative correction



1. Gate: no quantified RBF supplied → N/A; constants not recorded → UNKNOWN — honest, because an RBF map with undocumented constants is a number of unknown scale.
2. Compare each constant against its consensus value for the field strength and labelling scheme.
3. Compute the effective labelling efficiency: alpha_effective = alpha_nominal · 0.93 ** n_bs_pulses (R9.9) — PCASL with 3 pulses gives $0.85 \times 0.93^3 = 0.684$.
4. Compute the rescaling between the constants used and the consensus constants from the consensus's own single-compartment model (R9.2): f_lambda · f_alpha · f_t1_blood = total.
5. Report the factor and the below-consensus percentage as two separate numbers — and do not apply the correction; rescaling the user's map would be quantification, which is out of scope.

INPUT


```
{
  "rbf_supplied":   True,
  "labelling":      "PCASL",
  "field_T":        3.0,
  "constants_used": {
     "lambda":       0.8,      # mL/g
     "alpha":        0.85,     # labelling efficiency AS APPLIED
     "t1_blood_s":   1.55,
     "bgs_correction_applied": False,
     "n_bs_pulses":  3,
  ... 4 more lines
}
```


OUTPUT


```
{
  "metric": {
    "checked": {
      "lambda":     {"value": 0.8,  "recommended": 0.9,  "conforms": False,
                     "statement": "R9.4", "agreement": 90,  "abstention": 5},
      "t1_blood_s": {"value": 1.55, "recommended": 1.65, "conforms": False,
                     "statement": "R9.5", "agreement": 100, "abstention": 0},
      "bgs_alpha_correction": {"value": False, "recommended": True, "conforms": False,
                     "statement": "R9.9", "agreement": 100, "abstention": 19},
      # ... alpha (R9.8), model (R9.2)
  ... 12 more lines
}
```


VERDICT

PASS	every constant matches the consensus value for the field strength and labelling scheme, and the BGS α correction was applied where BS was used
WARN	any constant deviates — the reason states the total rescaling factor and its direction
WARN	background suppression was used but the α correction was omitted — a systematic underestimate, ~20% low for 3 pulses ($1 - 0.93^3$)
FAIL	never — a constant deviation is a documented, exactly-computable rescaling, not a corrupted measurement
N/A	no quantified RBF supplied (only ΔM)
UNKNOWN	quantification constants not recorded

All PUBLISHED, Nery 2020 — R9.4 λ = 0.9 mL/g (90%), R9.5 T1b = 1.65 s at 3 T (100%), R9.6 1.48 s at 1.5 T (93%), R9.7 α PASL 95% (100%), R9.8 α PCASL 85% (86%), R9.9 × 0.93 per BGS pulse (100%), R9.2 single-compartment single-delay model (100%).

λ = 0.9 is the consensus's own admitted brain-borrowed placeholder — "a reliable reference for the partition coefficient value in kidney was not known to this group" — which is the single strongest reason this check reports a factor instead of a verdict. And normalise the direction carefully: 1.289 is the upward correction; the map is 22.4% below consensus, not 28.9% off.